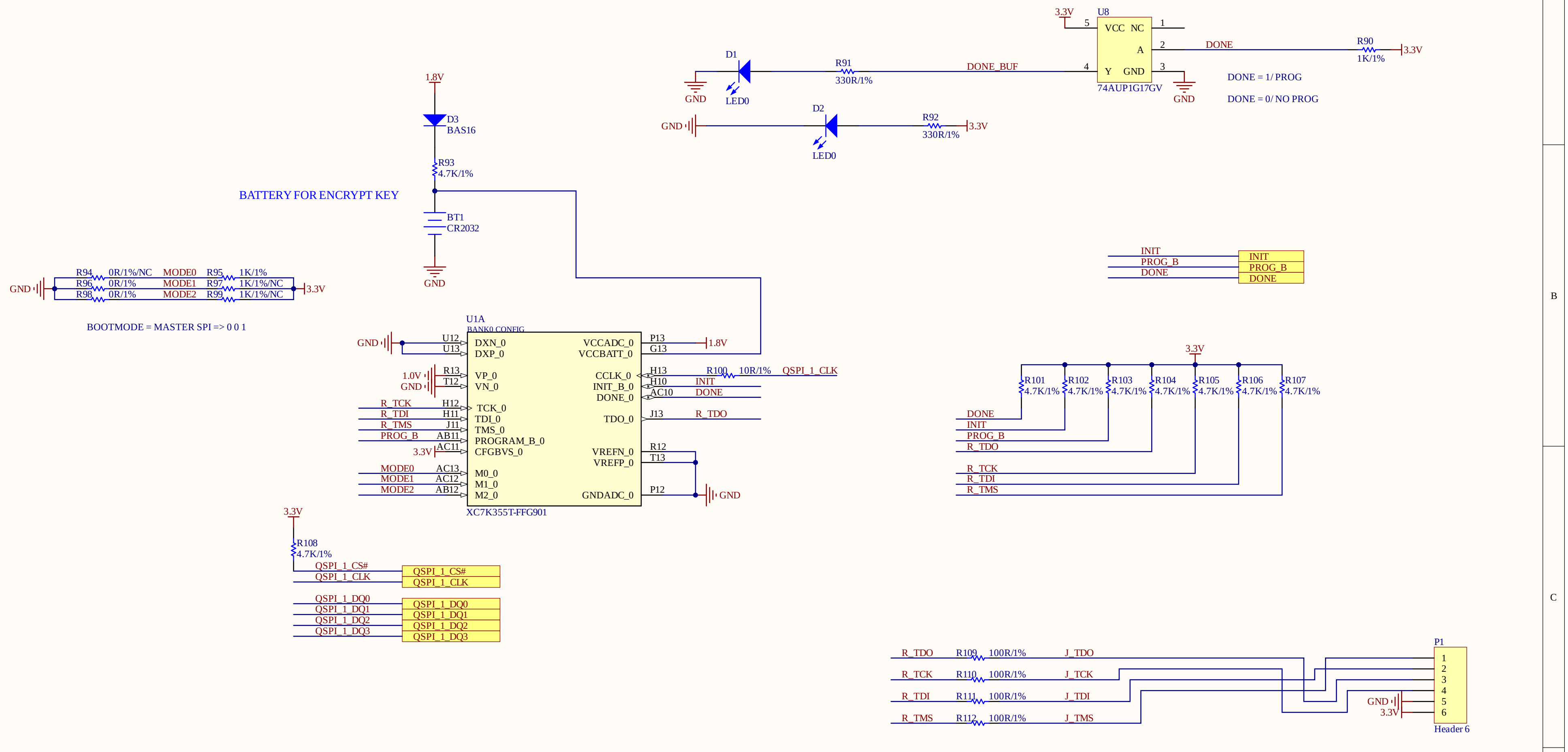
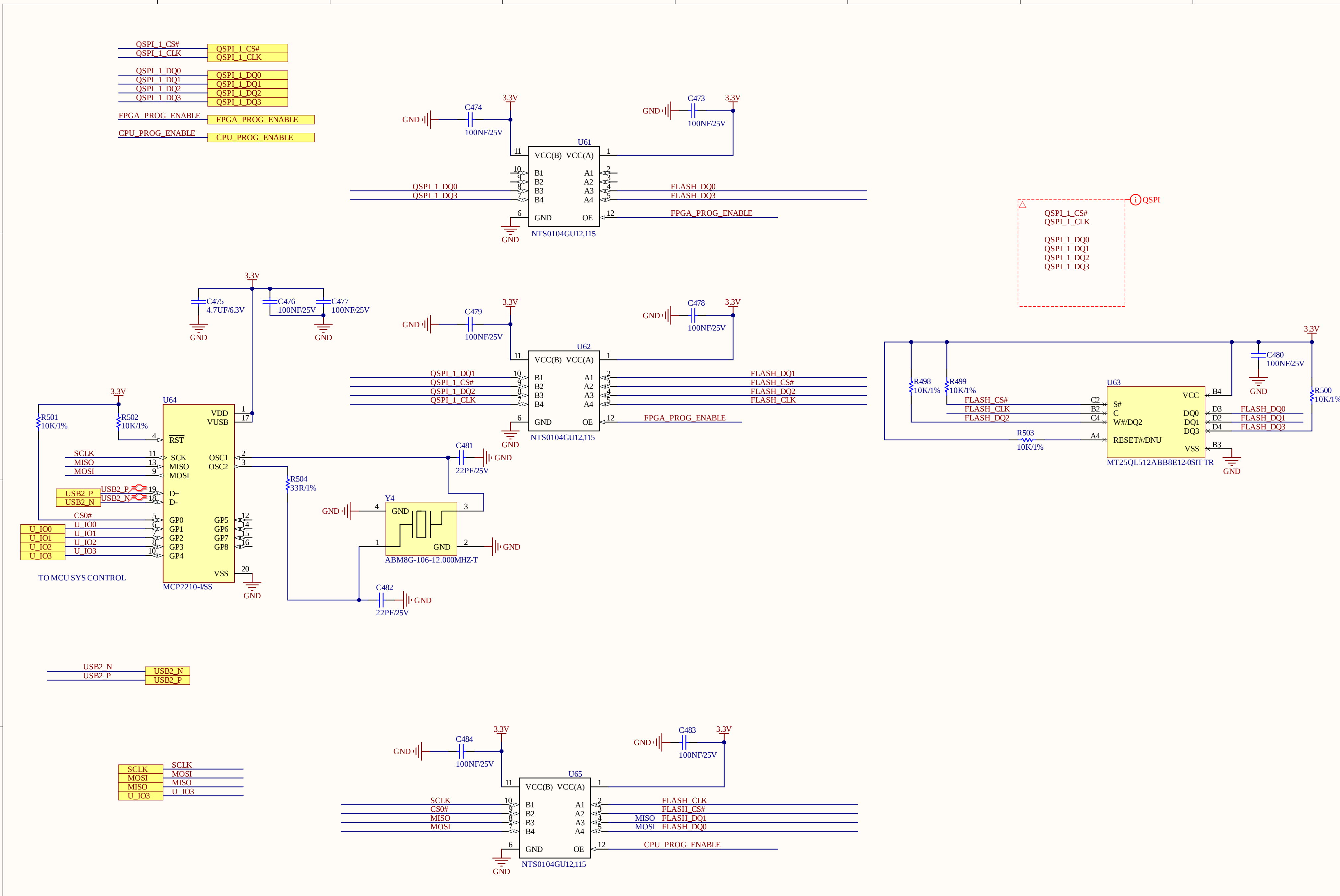


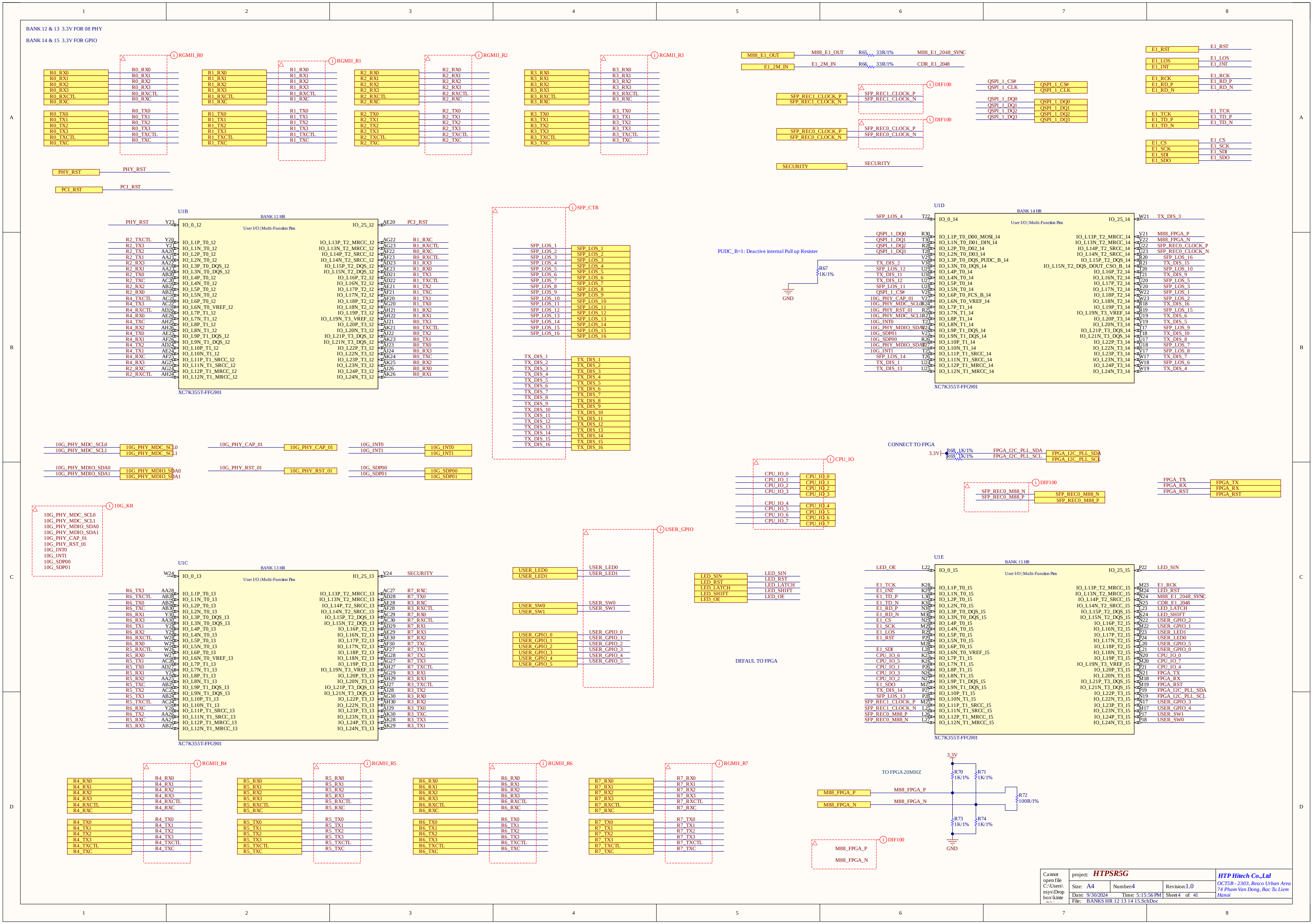
project: **HTPSR5G**			**HTP Hitech Co.,Ltd**
Size: A4	Number: 2	Revision: 1.0	*OCT5B - 2303, Resco Urban Area*
Date: 9/30/2024	Time: 5:15:55 PM	Sheet 2 of 41	*74 Pham Van Dong, Bac Tu Liem*
File: CONFIG.SchDoc			*Hanoi*

Cannot open file C:\Users\esys\Dropbox\kinte



Cannot open file C:\Users\esys\Dropbox\kinte	project: **HTPSR5G**		**HTP Hitech Co.,Ltd**
	Size: A4	Number: 3	Revision: 1.0
	Date: 9/30/2024	Time: 5:15:55 PM	Sheet 3 of 41
	File: SHARE FLASH.SchDoc		

OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



BANK 12 & 13 3.3V FOR 08 PHY
BANK 14 & 15 3.3V FOR GPIO
RGMII_R0
RGMII_R1
RGMII_R2
RGMII_R3
RGMII_R4
RGMII_R5
RGMII_R6
RGMII_R7
PHY_RST
PCI_RST
U1B
BANK 12 HR
User I/O | Multi-Function Pins
XC7K355T-FFG901
U1C
BANK 13 HR
U1D
BANK 14 HR
U1E
BANK 15 HR
SFP_CTR
DIF100
CPU_IO
USER_GPIO
10G_KR
PUDC_B=1: Deactive internal Pull up Resister
CONNECT TO FPGA
DEFAUL TO FPGA
TO FPGA 20MHZ
M88_FPGA_P
M88_FPGA_N
SECURITY
3.3V
GND
R65 33R/1%
R66 33R/1%
R67 1K/1%
R68 1K/1%
R69 1K/1%
R70 1K/1%
R71 1K/1%
R72 100R/1%
R73 1K/1%
R74 1K/1%
project: HTPSR5G
Size: A4
Number 4
Revision: 1.0
Date: 9/30/2024
Time: 5:15:56 PM
Sheet4 of 41
File: BANKS HR 12 13 14 15.SchDoc
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi

BANK 16 & 17   1.8V FOR QDR II

D and BW0/1 in same bank:0k
Only one write data byte is allowed per byte lane: 0k
All byte lanes that are used for the write data of a single component must be adjacent, no skip byte lanes are allowed: ok

One of the write data bytes of a memory component should be allocated in the: ok
Memory Clock pair must be allocated in one of the byte lanes that are used for the write data of the corresponding memory component.: ok

Memory clock should come from one of the center byte lanes (byte lanes 1 and 2).: ok
K/K# can be allocated to any PN pair: ok

All address/control (A/C) bits must be allocated in a single bank: ok
All A/C byte lanes should be contiguous and no skip byte lanes is allowed: ok
The address/control bank should be the same or adjacent to that of the write data bank: ok
There should not be any empty byte lane or read byte lane between A/C and write data byte lanes. This rule applies when A/C and write data share the same bank or allocated in adjacent banks: ok
Address/control pins should not share a byte lane with the write data as well as rea: ok

The entire read data bus must be placed in a single bank irrespective of the number of memory components.: ok

All byte lanes that are used for the read data of a single component must be adjacent, no skip byte lanes are allowed.: ok

One of the read data bytes of a memory component should be allocated in the center byte lanes (byte lanes 1 and 2).: ok

If a byte lane is used for read data, Bit[0] and Bit[6] must be used. Read clock (CQ or CQ#) gets the first priority and data (Q) is the next.: ok

Read data buses of two components should not share a byte lane.: ok

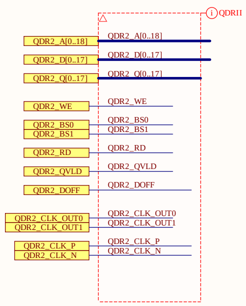
Read Clock pair must be allocated in one of the byte lanes that are used for the read: ok

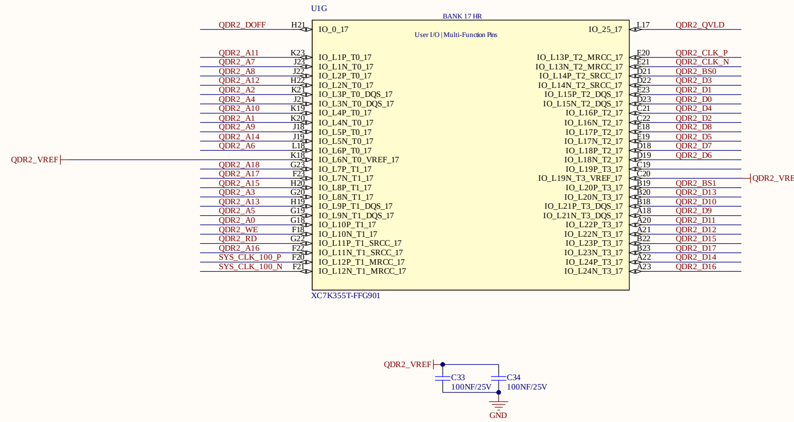
CQ/CQ# pair must be allocated in a single byte lane.: ok

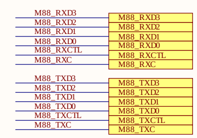
CQ/CQ# must be allocated only in the byte lanes 1 and 2 because the other byte: ok

CQ/CQ# must be allocated in the center byte lane of all used byte lanes. If two byte lanes are used for the read data, either one of them can be used for CQ/CQ# allocation: ok

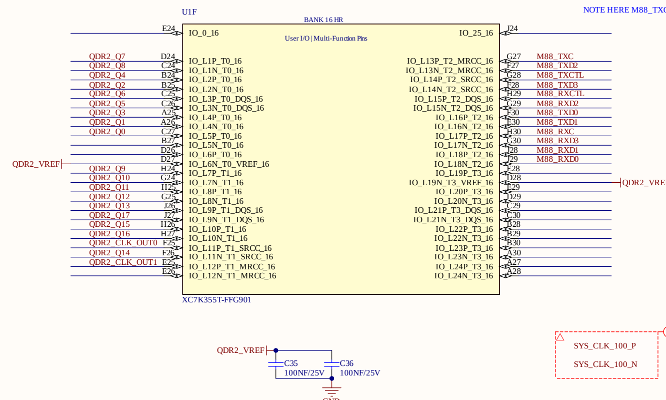
CQ and CQ# must be allocated to either pin 0 or pin 6 of a byte lane. For example, if CQ is allocated to pin 0, CQ# should be allocated to pin 6 and vice versa: ok



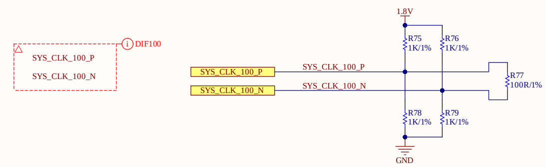




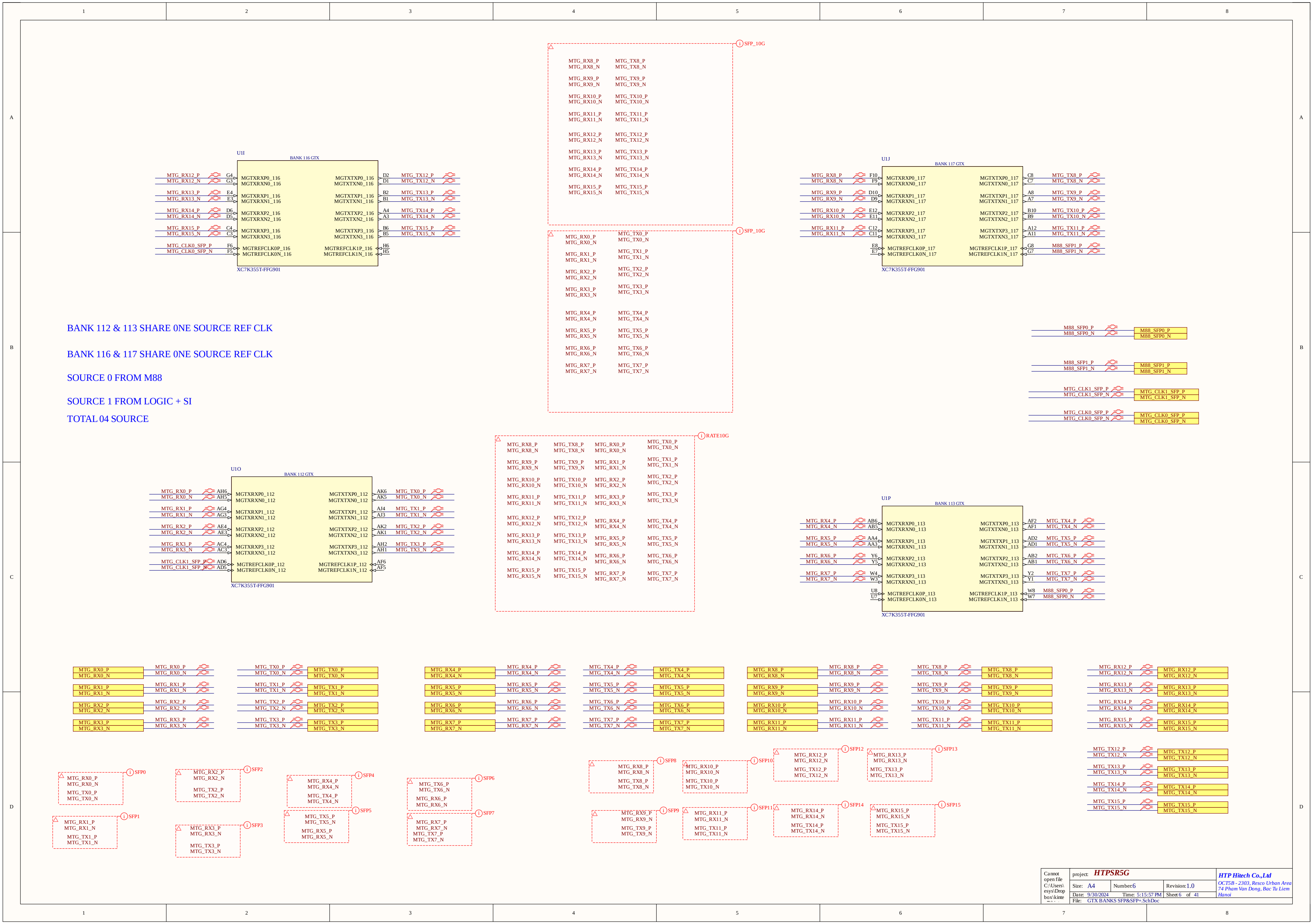
NOTE HERE M88_TXC NEED CLOCK FUNCTION



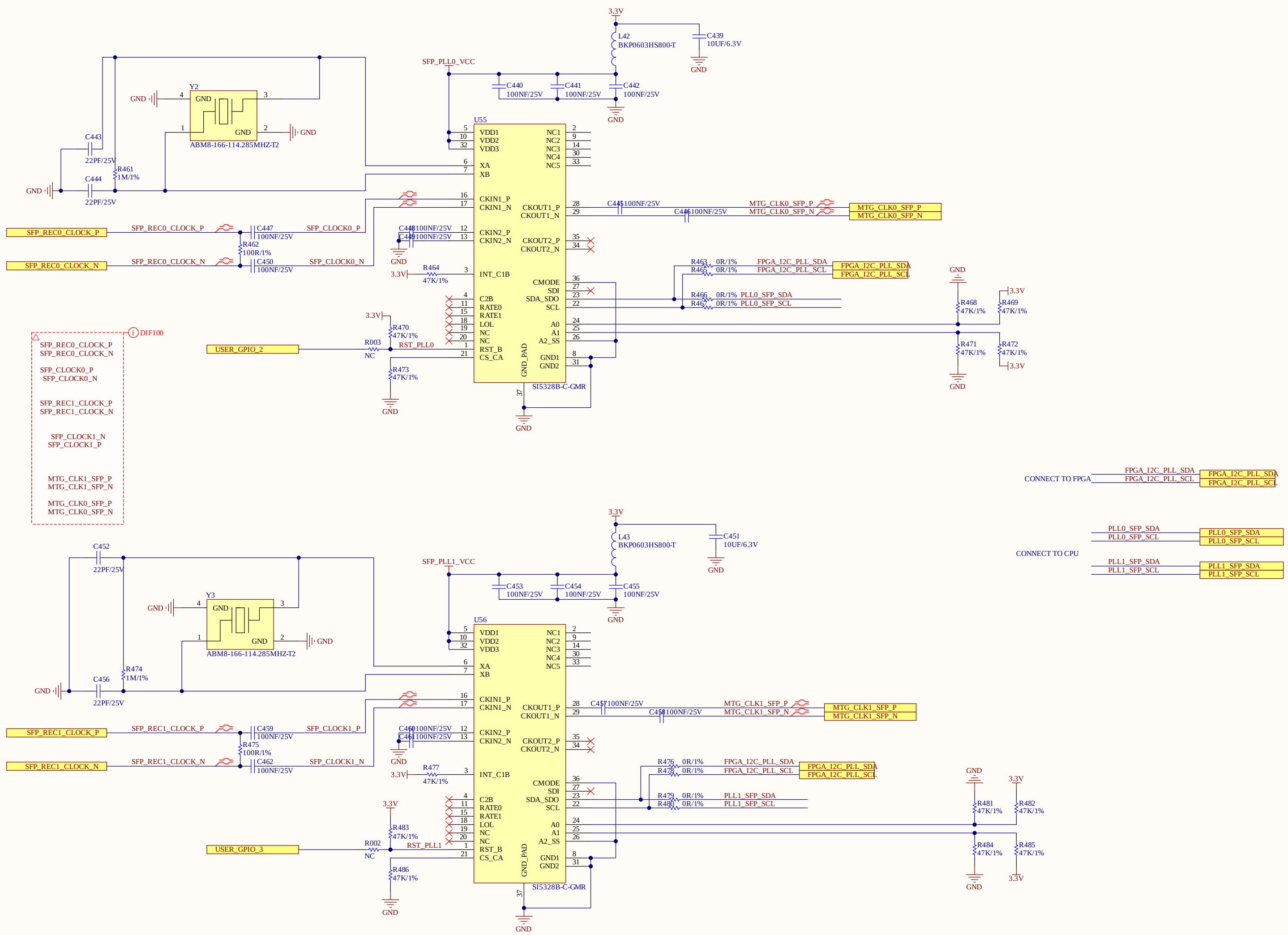
Pins can swap freely within each Write Data byte group.
• Pins can swap freely within each Read Data byte group, except CQ/CQ# pins. Pins can swap freely within and between their corresponding byte groups, but should not violate above mentioned Read Data pin/byte lane allocation rules.
• Pins can swap freely withing each Address/Control byte group. Pins can swap freely within and between their corresponding byte groups, but should not violate above mentioned Address/Control pin/byte lane allocation rules.
• Write Data Byte groups can swap easily with each other, but should not violate above mentioned Write Data pin/byte lane allocation rules.
• Read Data Byte groups can swap easily with each other, but should not violate above mentioned Read Data pin/byte lane allocation rules.
• Address/Control Byte groups can swap easily with each other, but should not violate above mentioned Address/Control pin/byte lane allocation rules.
• No other pin swapping is permitted.



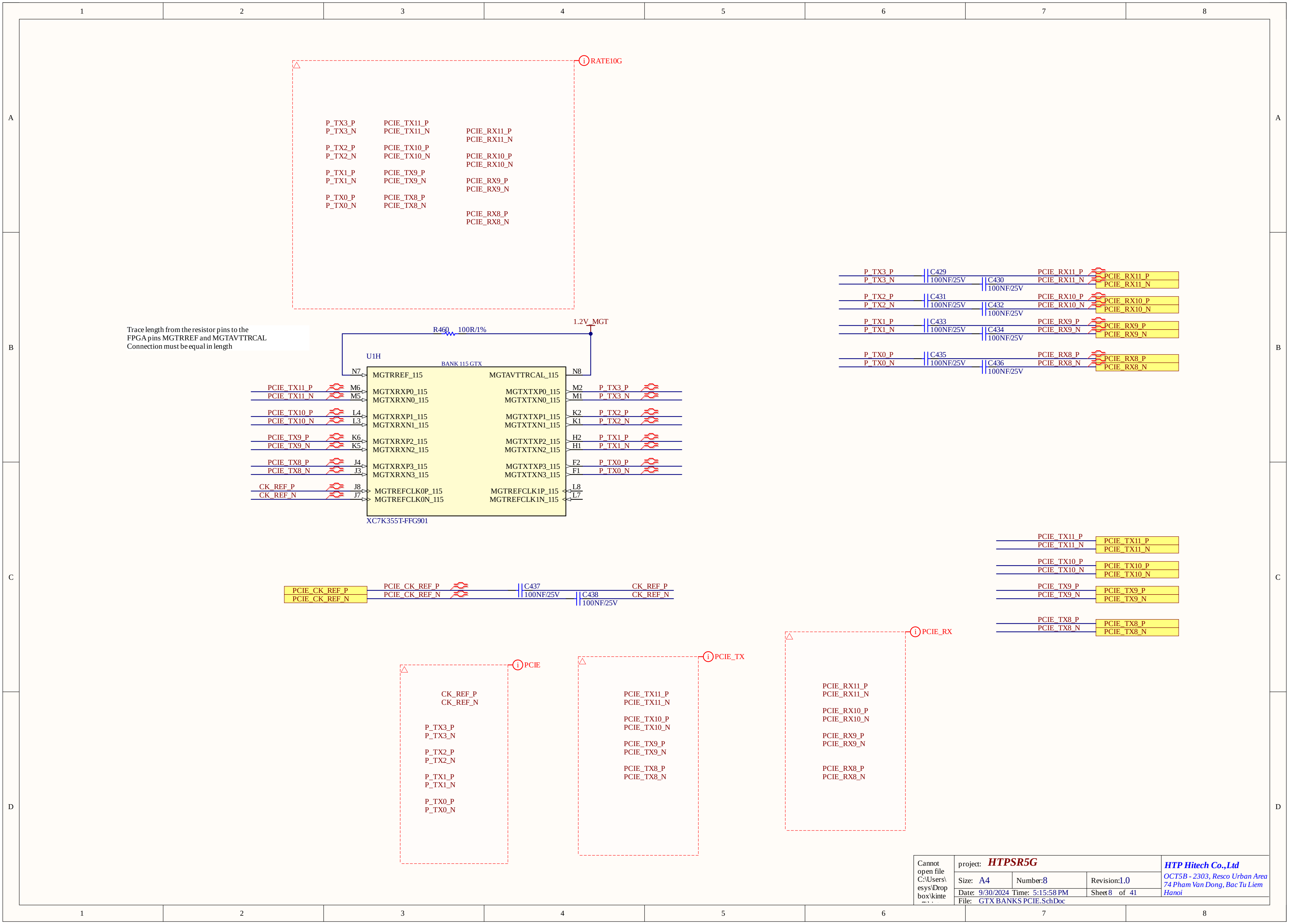
Cannot open file C:\Users\esys\Dropbox\kinte	project: **HTPSR5G**		**HTP Hitech Co.,Ltd**
	Size: A4	Number: 5	Revision: 1.0
	Date: 9/30/2024	Time: 5:15:56 PM	Sheet 5 of 41
	File: BANKS HR 16 17.SchDoc		*OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi*



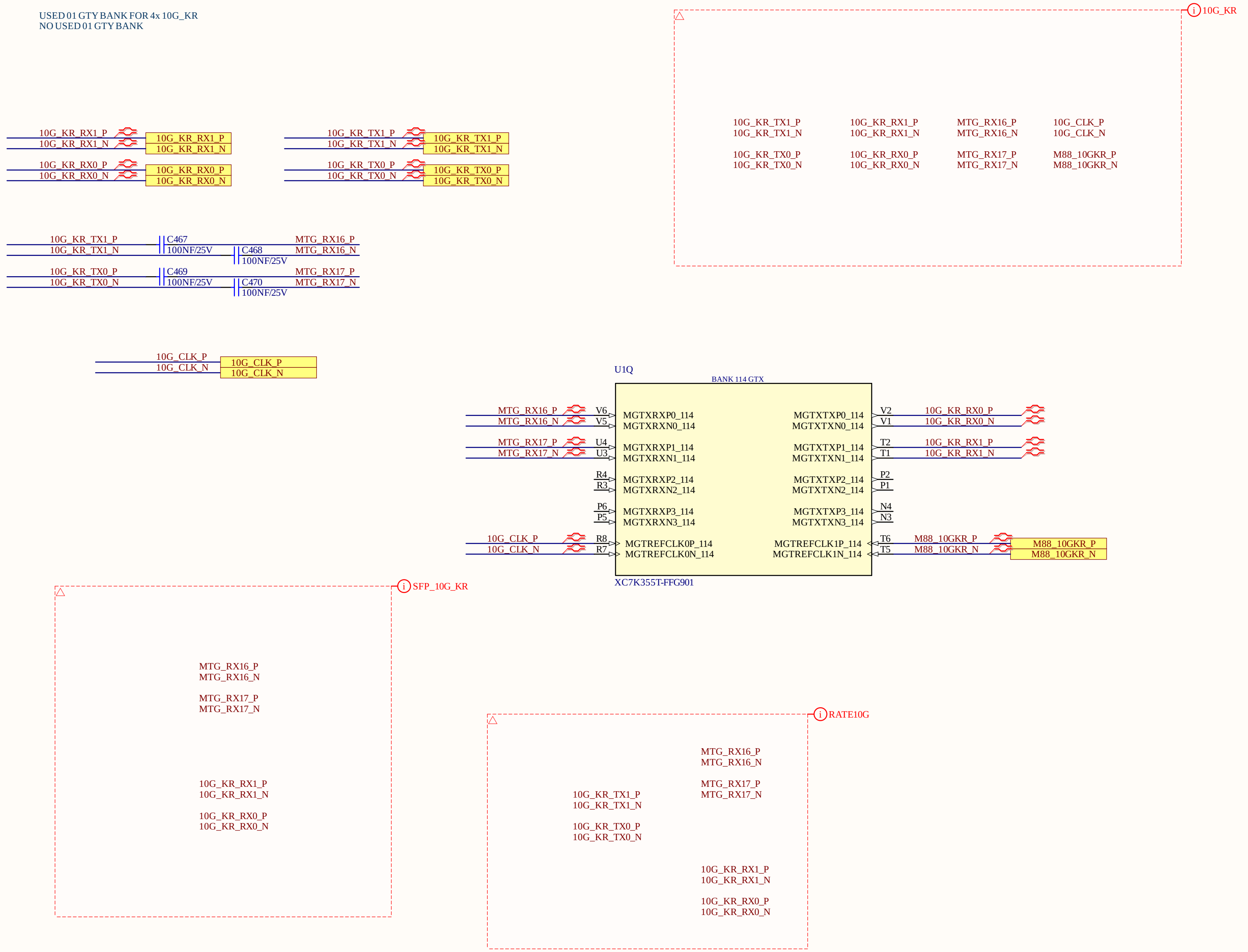
BANK 112 & 113 SHARE 0NE SOURCE REF CLK
BANK 116 & 117 SHARE 0NE SOURCE REF CLK
SOURCE 0 FROM M88
SOURCE 1 FROM LOGIC + SI
TOTAL 04 SOURCE
U1I BANK 116 GTX XC7K355T-FFG901
U1J BANK 117 GTX XC7K355T-FFG901
U1O BANK 112 GTX XC7K355T-FFG901
U1P BANK 113 GTX XC7K355T-FFG901
SFP_10G
SFP_10G
RATE10G
SFP0 SFP1 SFP2 SFP3 SFP4 SFP5 SFP6 SFP7 SFP8 SFP9 SFP10 SFP11 SFP12 SFP13 SFP14 SFP15
project: HTPSR5G
Size: A4
Number:6
Revision:1.0
Date: 9/30/2024
Time: 5:15:57 PM
Sheet6 of 41
File: GTX BANKS SFP&SFP+.SchDoc
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



Cannot open file C:\Users\esys\Drop box\kinte	project: **HTPSR5G**	**HTP Hitech Co.,Ltd**
	Size: A4 Number: 7 Revision: 1.0	*OCT5B - 2303, Resco Urban Area*
	Date: 9/30/2024 Time: 5:15:57 PM Sheet 7 of 41	*74 Pham Van Dong, Bac Tu Liem*
	File: GTX SFP REF CLKS.SchDoc	*Hanoi*


RATE10G
P_TX3_P
P_TX3_N
P_TX2_P
P_TX2_N
P_TX1_P
P_TX1_N
P_TX0_P
P_TX0_N
PCIE_TX11_P
PCIE_TX11_N
PCIE_TX10_P
PCIE_TX10_N
PCIE_TX9_P
PCIE_TX9_N
PCIE_TX8_P
PCIE_TX8_N
PCIE_RX11_P
PCIE_RX11_N
PCIE_RX10_P
PCIE_RX10_N
PCIE_RX9_P
PCIE_RX9_N
PCIE_RX8_P
PCIE_RX8_N
Trace length from the resistor pins to the
FPGA pins MGTRREF and MGTAVTTRCAL
Connection must be equal in length
R460 100R/1%
1.2V_MGT
U1H
BANK 115 GTX
N7 MGTRREF_115
MGTAVTTRCAL_115 N8
M6 MGTXRXP0_115
M5 MGTXRXN0_115
L4 MGTXRXP1_115
L3 MGTXRXN1_115
K6 MGTXRXP2_115
K5 MGTXRXN2_115
J4 MGTXRXP3_115
J3 MGTXRXN3_115
J8 MGTREFCLK0P_115
J7 MGTREFCLK0N_115
MGTXTXP0_115 M2
MGTXTXN0_115 M1
MGTXTXP1_115 K2
MGTXTXN1_115 K1
MGTXTXP2_115 H2
MGTXTXN2_115 H1
MGTXTXP3_115 F2
MGTXTXN3_115 F1
MGTREFCLK1P_115 L8
MGTREFCLK1N_115 L7
XC7K355T-FFG901
CK_REF_P
CK_REF_N
C429 100NF/25V
C430 100NF/25V
C431 100NF/25V
C432 100NF/25V
C433 100NF/25V
C434 100NF/25V
C435 100NF/25V
C436 100NF/25V
PCIE_CK_REF_P
PCIE_CK_REF_N
C437 100NF/25V
C438 100NF/25V
PCIE
PCIE_TX
PCIE_RX
Cannot open file C:\Users\esys\Drop box\kinte
project: HTPSR5G
Size: A4
Number: 8
Revision: 1.0
Date: 9/30/2024
Time: 5:15:58 PM
Sheet 8 of 41
File: GTX BANKS PCIE.SchDoc
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi

USED 01 GTY BANK FOR 4x 10G_KR
NO USED 01 GTY BANK



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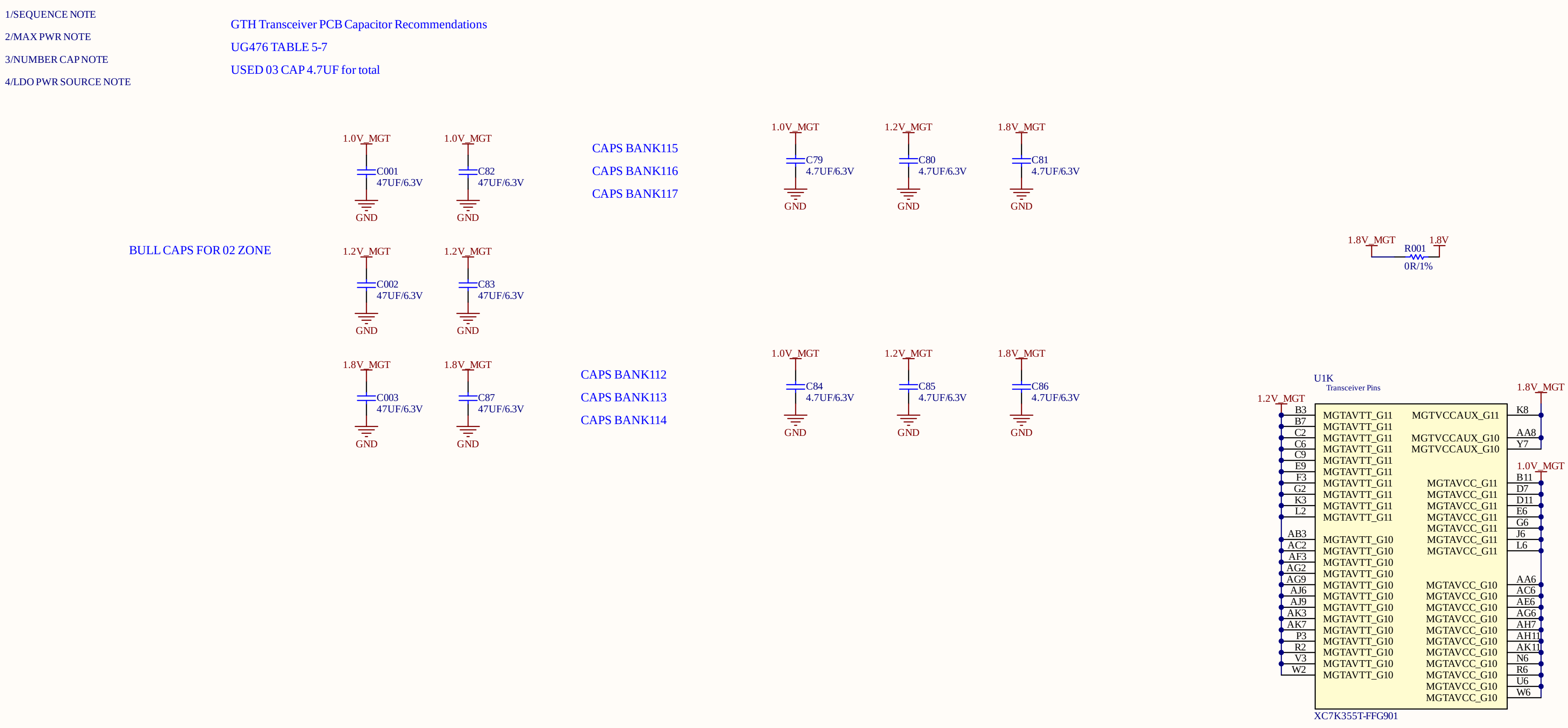
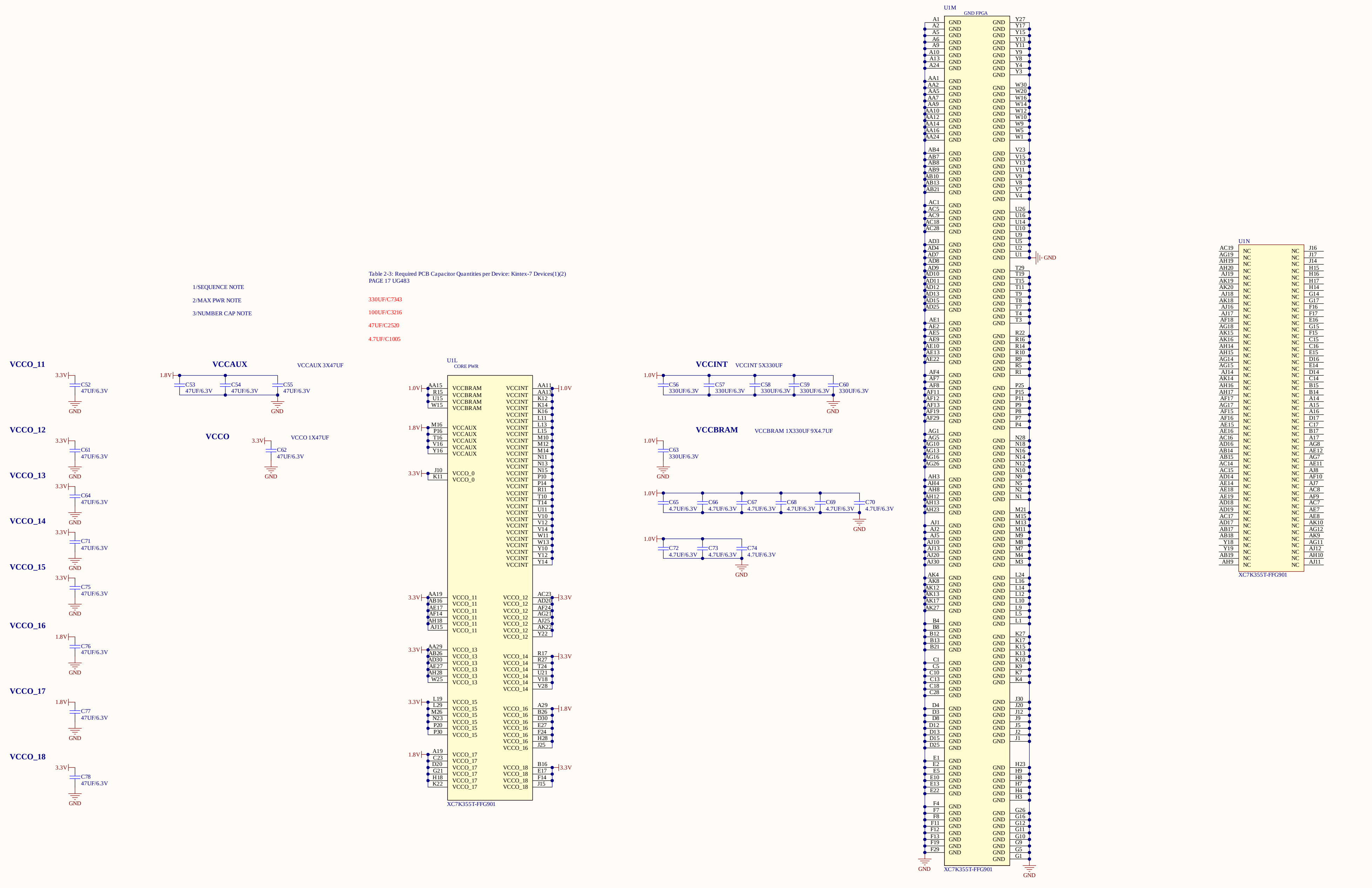
	Size: A4	Number: 9	Revision: 1.0
	Date: 9/30/2024 Time: 5:15:58 PM	Sheet 9 of 41	*OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi*
	File: 10G KR INTERFACE.SchDoc		



Cannot open file C:\Users\esys\Dropbox\kinte	project: ***HTPSR5G***		**HTP Hitech Co.,Ltd** *OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi*
	Size: A4	Number: 10	Revision: 1.0
	Date: 9/30/2024 Time: 5:15:58 PM		Sheet 10 of 41
	File: GTX PWR.SchDoc		



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	Size: A4	Number: 11	Revision: 1.0
	Date: 9/30/2024 Time: 5:15:59 PM		Sheet 11 of 41
	File: FPGA PWR.SchDoc		OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi

PWR SEQUENCE FPGA:

1/ VCCINT/VCCBRAM: 1.0V

2/ VCCAUX,VCCAUX_IO,VCCO HP BANK: 1.8V

3/ VCCO HR BANK: 3.3V

PWR RATE FPGA:

1/ VCCINT: 1.0V/40A

2/ VCCAUX,VCCAUX_IO,VCCO HP BANK: 1.8V/10A

3/ VCCO HR BANK: 3.3V/10A

PWR SEQUENCE GTX:

1/ VCCINT: 1.0V

2/ VMGTAVCC: 1V

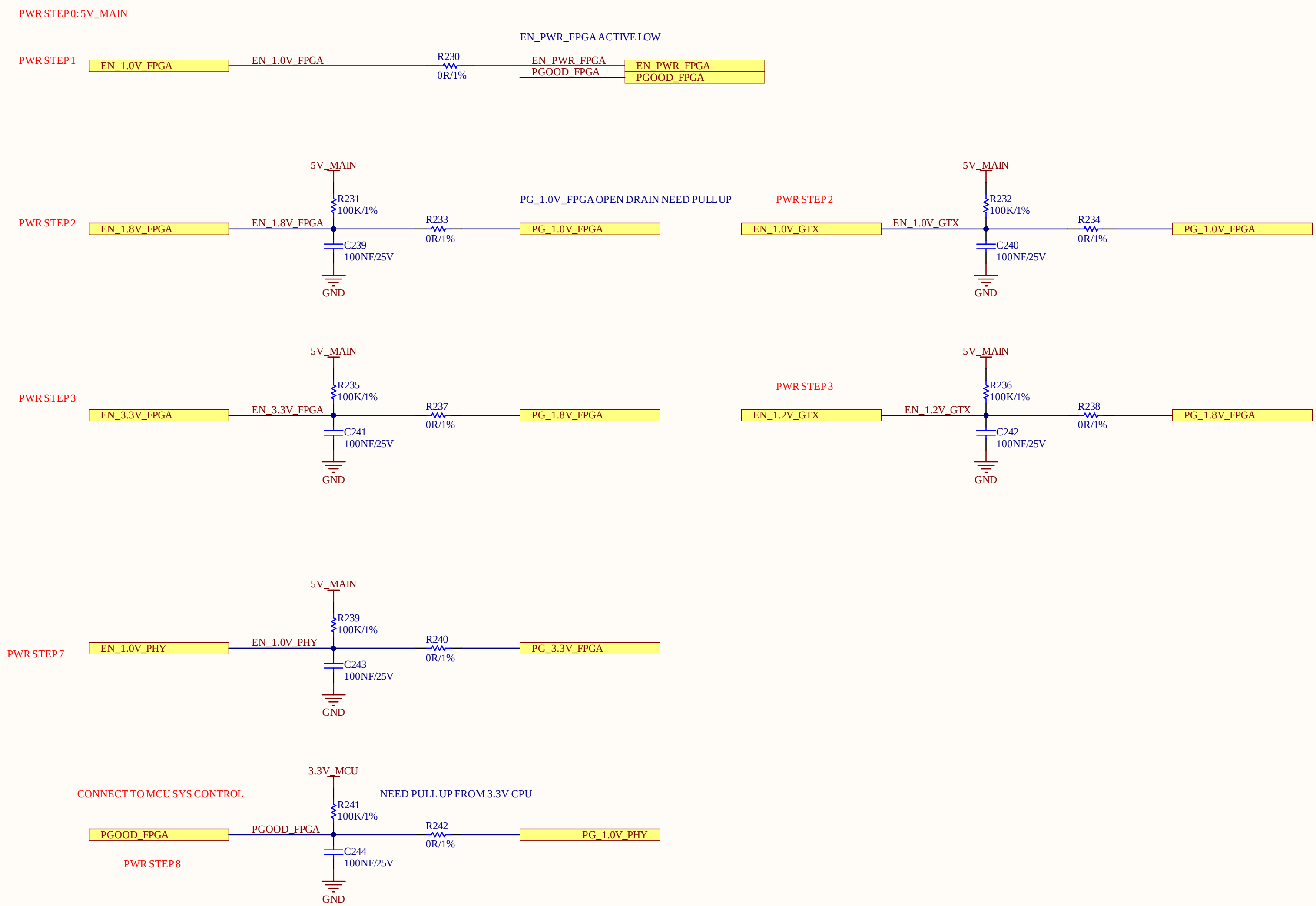
3/ VMGTAVTT: 1.2V & VMGTVCCAUX: 1.8V

PWR RATE GTX:

1/ VMGTAVCC: 1V/8A

2/ MGTVCCAUX: 1.8V SHARE

3/ VMGTAVTT: 1.2V/8A



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	Size: A4	Number: 12	Revision: 1.0
	Date: 9/30/2024	Time: 5:15:59 PM	Sheet 12 of 41
	File: PWR SEQ.SchDoc		

HTP Hitech Co.,Ltd

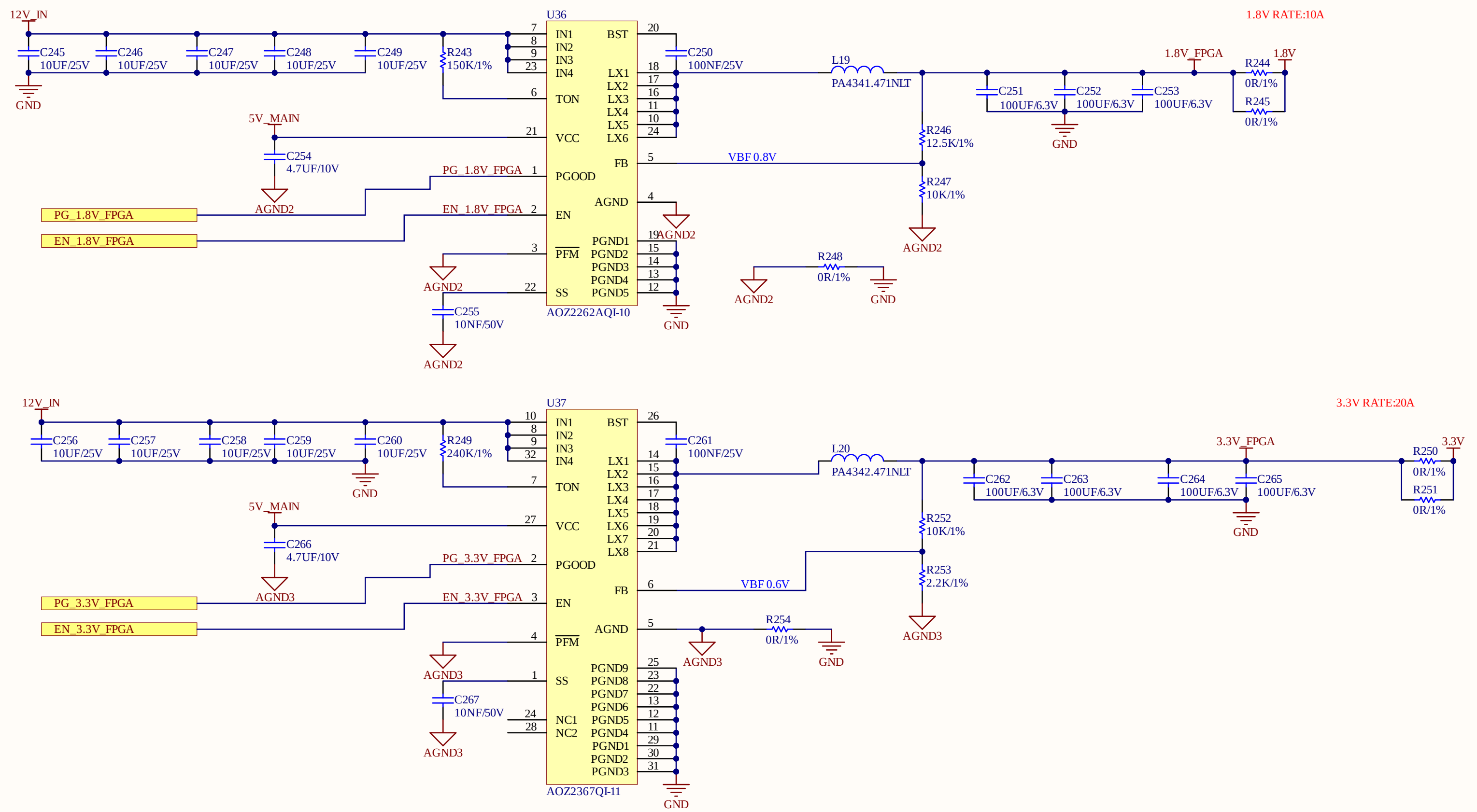
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi

PWR SEQUENCE FPGA:

1/ VCCINT/VCCBRAM: 1.0V
2/ VCCAUX,VCCAUX_IO,VCCO HP BANK: 1.8V
3/ VCCO HR BANK: 3.3V

PWR RATE FPGA:

1/ VCCINT: 1.0V/50A
2/ VCCAUX,VCCAUX_IO,VCCO HP BANK: 1.8V/10A
3/ VCCO HR BANK: 3.3V/20A



PWR SEQUENCE PHY:

1/ 1.05V_PHY

2/ 3.3V_PHY

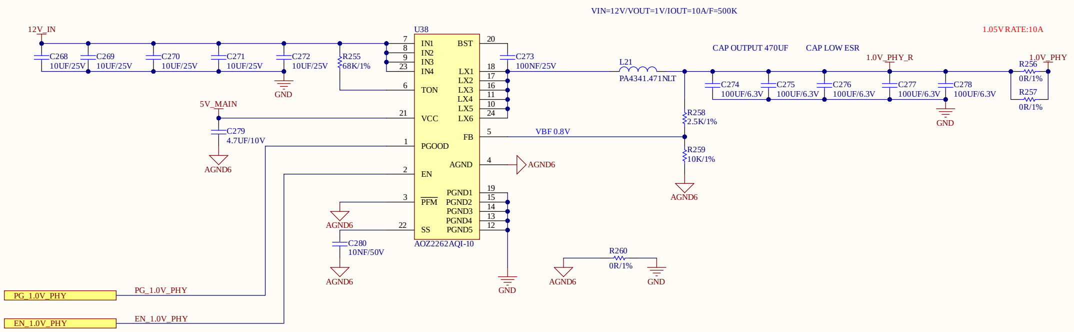
PWR RATE FPGA:

3/ 3.3V_PHY SHARE 3.3V

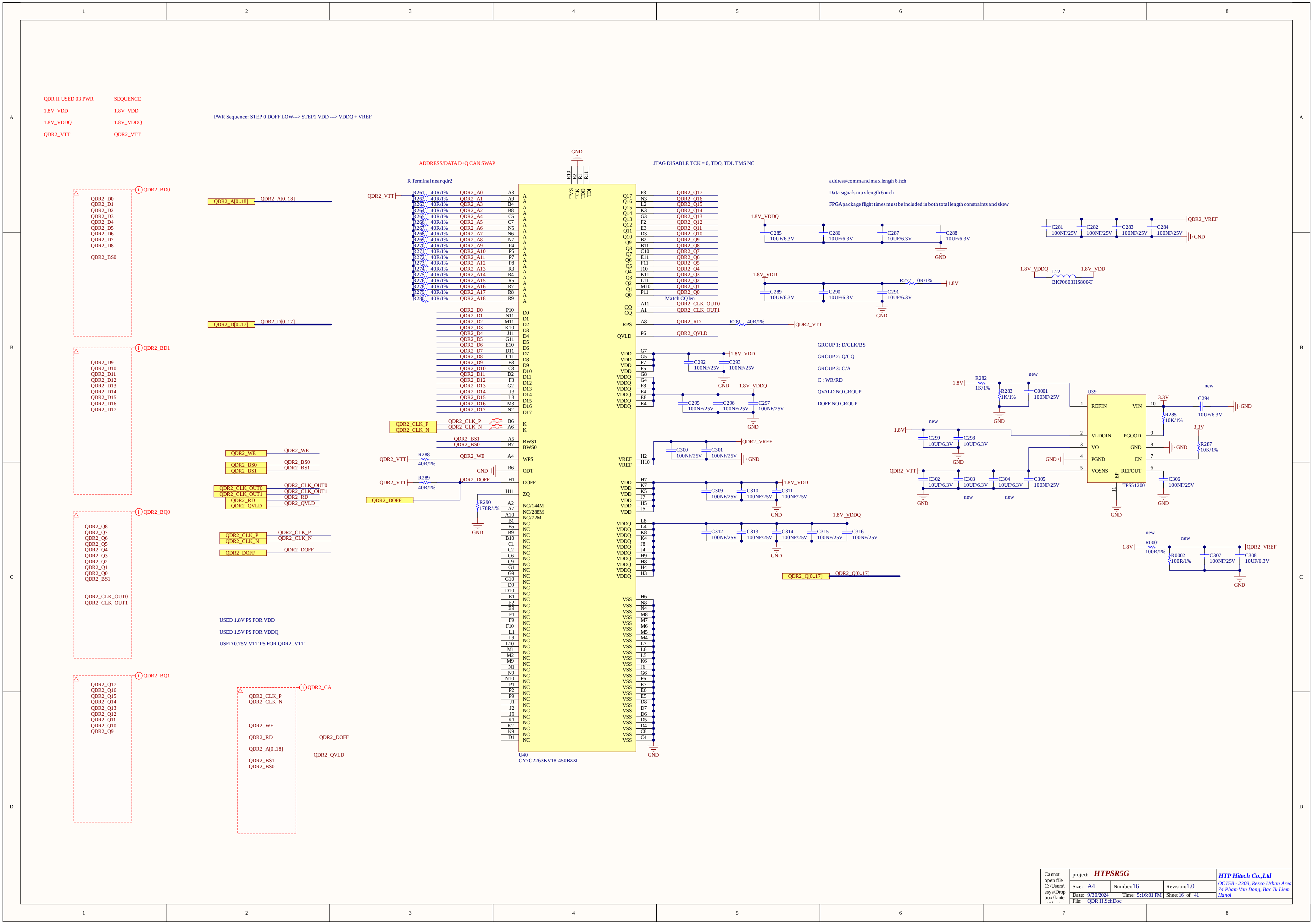
2/ 1.05V_PHY/5A PHY 1.05V RATE 5A, 8X0.6=4.8A

3.3V_PHY SHARE 3.3V SYSTEM

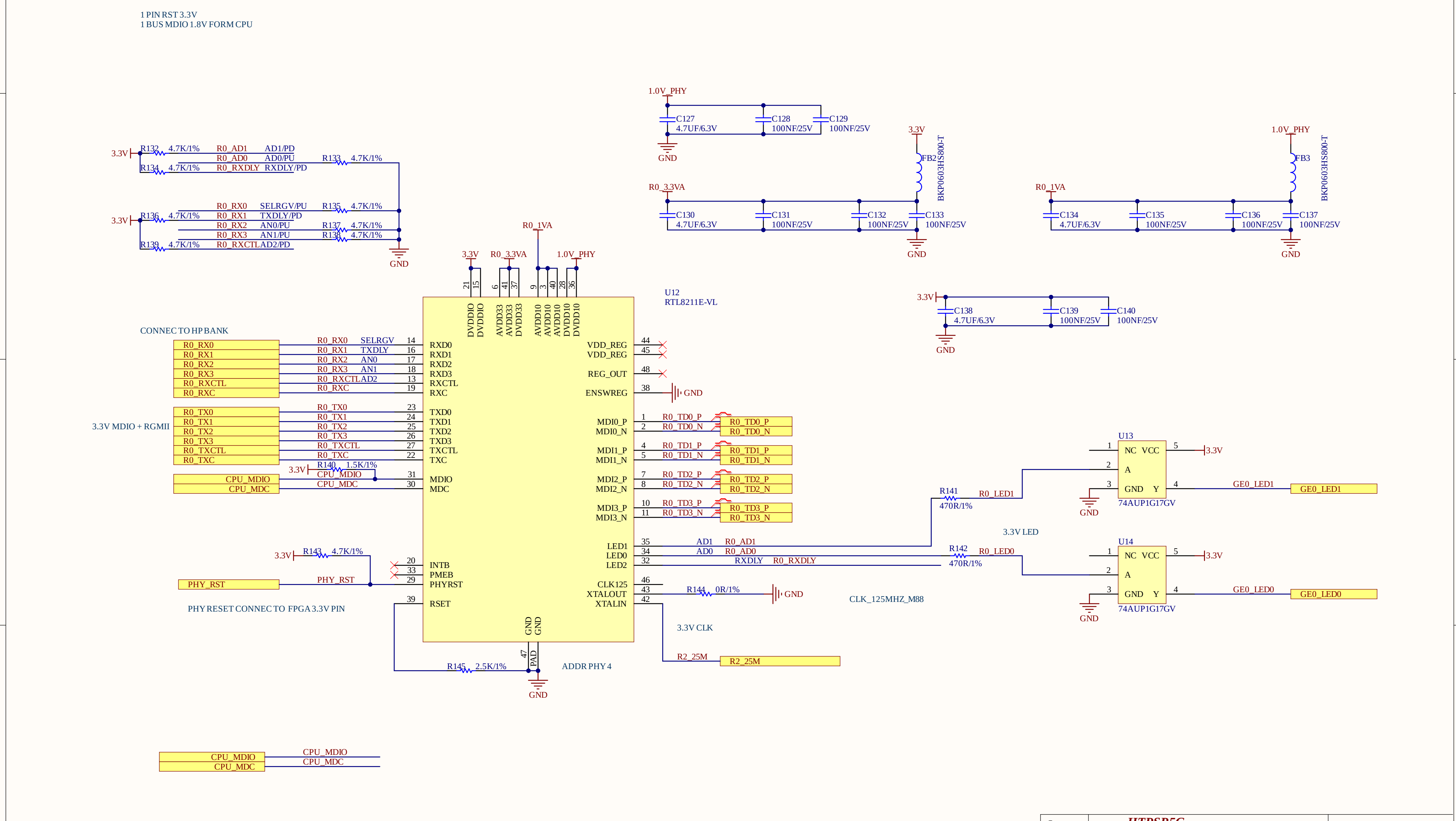
1.0V_PHY RATE 5A



Cannot open file C:\Users\esys\Dropbox\kinte	project: **HTPSR5G**		**HTP Hitech Co.,Ltd** OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi
	Size: A4	Number: 15	Revision: 1.0
	Date: 9/30/2024 Time: 5:16:00 PM	Sheet 15 of 41	
	File: PWR PHY.SchDoc		



QDR II USED 03 PWR
SEQUENCE
1.8V_VDD
1.8V_VDD
1.8V_VDDQ
1.8V_VDDQ
QDR2_VTT
QDR2_VTT
PWR Sequence: STEP 0 DOFF LOW---> STEP1 VDD ---> VDDQ + VREF
ADDRESS/DATA D+Q CAN SWAP
JTAG DISABLE TCK = 0, TDO, TDI. TMS NC
R Terminal near qdr2
address/command max length 6 inch
Data signals max length 6 inch
FPGA package flight times must be included in both total length constraints and skew
Match CQ len
GROUP 1: D/CLK/BS
GROUP 2: Q/CQ
GROUP 3: C/A
C : WR/RD
QVALD NO GROUP
DOFF NO GROUP
USED 1.8V PS FOR VDD
USED 1.5V PS FOR VDDQ
USED 0.75V VTT PS FOR QDR2_VTT
QDR2_BD0
QDR2_BD1
QDR2_BQ0
QDR2_BQ1
QDR2_CA
QDR2_A[0..18]
QDR2_D[0..17]
QDR2_Q[0..17]
QDR2_CLK_P
QDR2_CLK_N
QDR2_WE
QDR2_BS0
QDR2_BS1
QDR2_CLK_OUT0
QDR2_CLK_OUT1
QDR2_RD
QDR2_QVLD
QDR2_DOFF
QDR2_VREF
U40
CY7C2263KV18-450BZXI
U39
TPS51200
L22
BKP0603HS800-T
Cannot open file C:\Users\esys\Dropbox\kinte
project: HTPSR5G
Size: A4
Number: 16
Revision: 1.0
Date: 9/30/2024
Time: 5:16:01 PM
Sheet 16 of 41
File: QDR II.SchDoc
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



Cannot open file C:\Users\esys\Dropbox\kinte	project: HTPSR5G		HTP Hitech Co.,Ltd
	Size: A4	Number: 17	Revision: 1.0
	Date: 9/30/2024	Time: 5:16:01 PM	Sheet 17 of 41
	File: PHY0.SchDoc		

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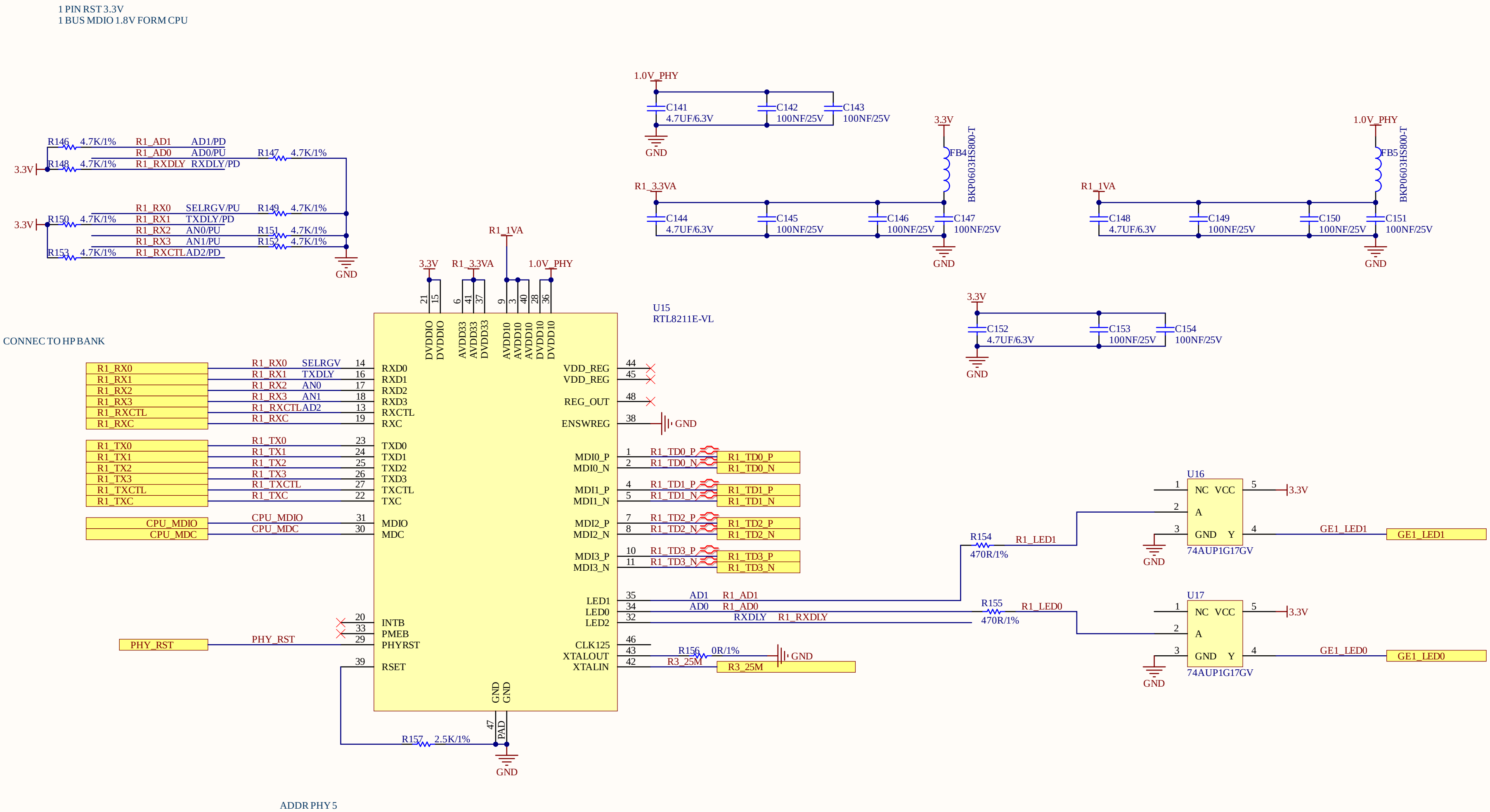
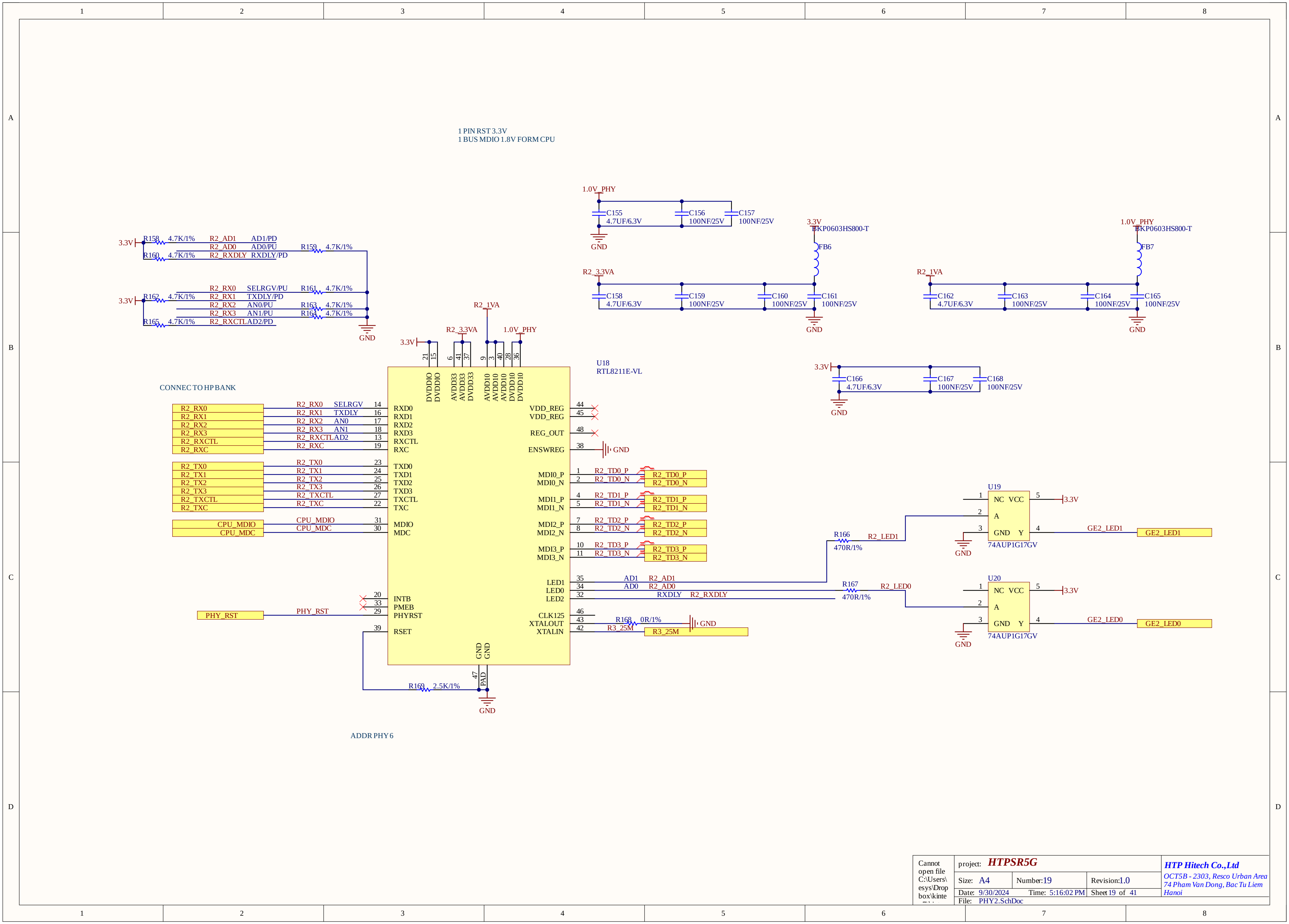
74 Pham Van Dong, Bac Tu Liem
Hanoi



Cannot open file C:\Users\esys\Dropbox\kinte	project: ***HTPSR5G***		
	Size: A4	Number: 18	Revision: 1.0
	Date: 9/30/2024	Time: 5:16:01 PM	Sheet 18 of 41
	File: PHY1.SchDoc		

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi


1 PIN RST 3.3V
1 BUS MDIO 1.8V FORM CPU
CONNECT TO HP BANK
ADDR PHY 6
U18
RTL8211E-VL
U19
74AUP1G17GV
U20
74AUP1G17GV
R2_RX0
R2_RX1
R2_RX2
R2_RX3
R2_RXCTL
R2_RXC
R2_TX0
R2_TX1
R2_TX2
R2_TX3
R2_TXCTL
R2_TXC
CPU_MDIO
CPU_MDC
PHY_RST
R2_TD0_P
R2_TD0_N
R2_TD1_P
R2_TD1_N
R2_TD2_P
R2_TD2_N
R2_TD3_P
R2_TD3_N
R3_25M
GE2_LED1
GE2_LED0
R2_LED1
R2_LED0
R2_AD1
R2_AD0
R2_RXDLY
SELRGV
TXDLY
AN0
AN1
AD2
AD1/PD
AD0/PU
RXDLY/PD
SELRGV/PU
TXDLY/PD
AN0/PU
AN1/PU
AD2/PD
R158 4.7K/1%
R159 4.7K/1%
R160 4.7K/1%
R161 4.7K/1%
R162 4.7K/1%
R163 4.7K/1%
R164 4.7K/1%
R165 4.7K/1%
R166 470R/1%
R167 470R/1%
R168 0R/1%
R169 2.5K/1%
C155 4.7UF/6.3V
C156 100NF/25V
C157 100NF/25V
C158 4.7UF/6.3V
C159 100NF/25V
C160 100NF/25V
C161 100NF/25V
C162 4.7UF/6.3V
C163 100NF/25V
C164 100NF/25V
C165 100NF/25V
C166 4.7UF/6.3V
C167 100NF/25V
C168 100NF/25V
FB6
FB7
BKP0603HS800-T
1.0V_PHY
R2_3.3VA
R2_1VA
3.3V
GND
RXD0
RXD1
RXD2
RXD3
RXCTL
RXC
TXD0
TXD1
TXD2
TXD3
TXCTL
TXC
MDIO
MDC
INTB
PMEB
PHYRST
RSET
DVDDIO
AVDD33
AVDD10
DVDD10
VDD_REG
REG_OUT
ENSWREG
MDI0_P
MDI0_N
MDI1_P
MDI1_N
MDI2_P
MDI2_N
MDI3_P
MDI3_N
LED1
LED0
LED2
CLK125
XTALOUT
XTALIN
PAD
NC
VCC
A
Y
Cannot open file C:\Users\esys\Dropbox\kinte
project: HTPSR5G
Size: A4
Number: 19
Revision: 1.0
Date: 9/30/2024
Time: 5:16:02 PM
Sheet 19 of 41
File: PHY2.SchDoc
HTP Hitech Co.,Ltd
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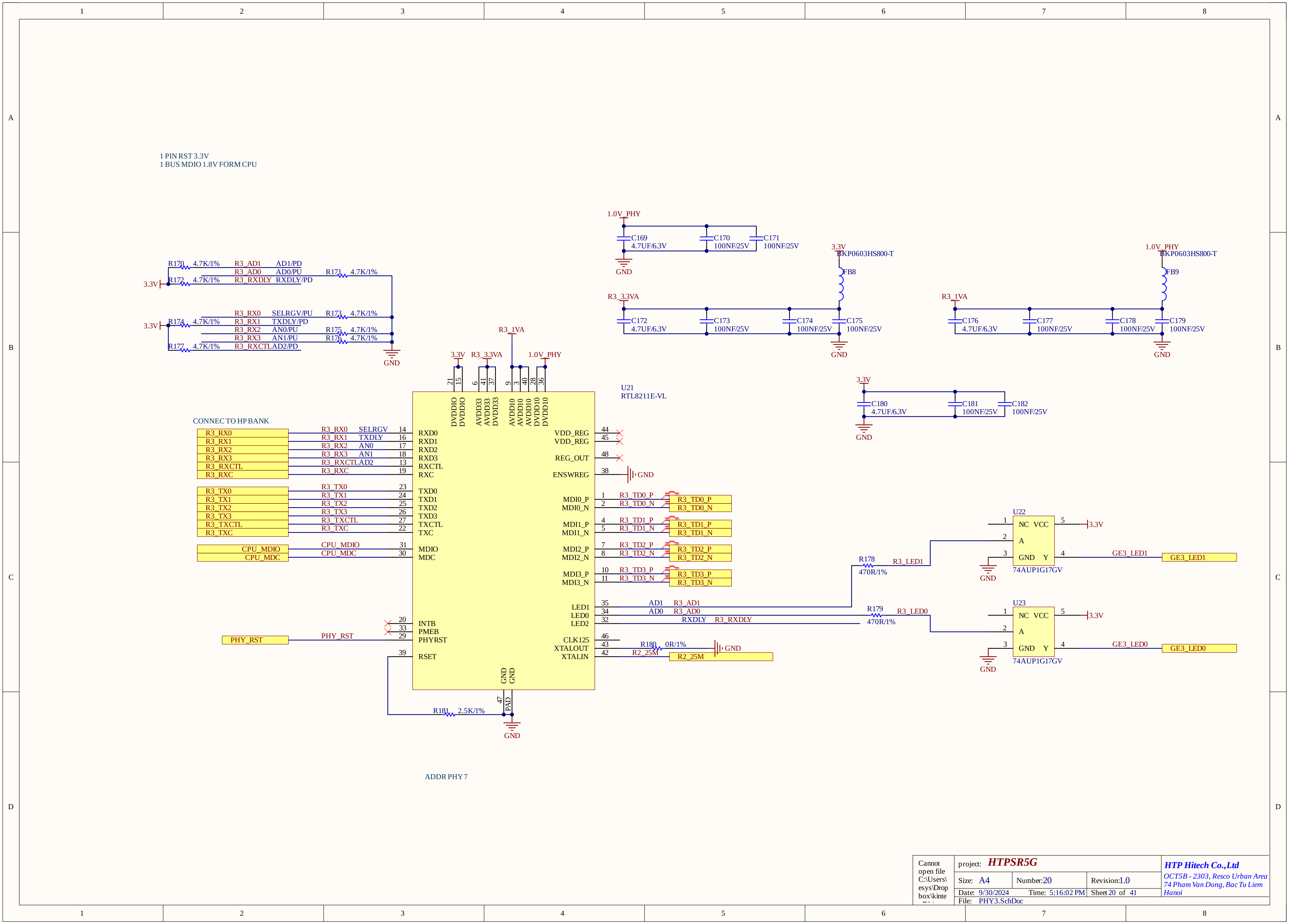
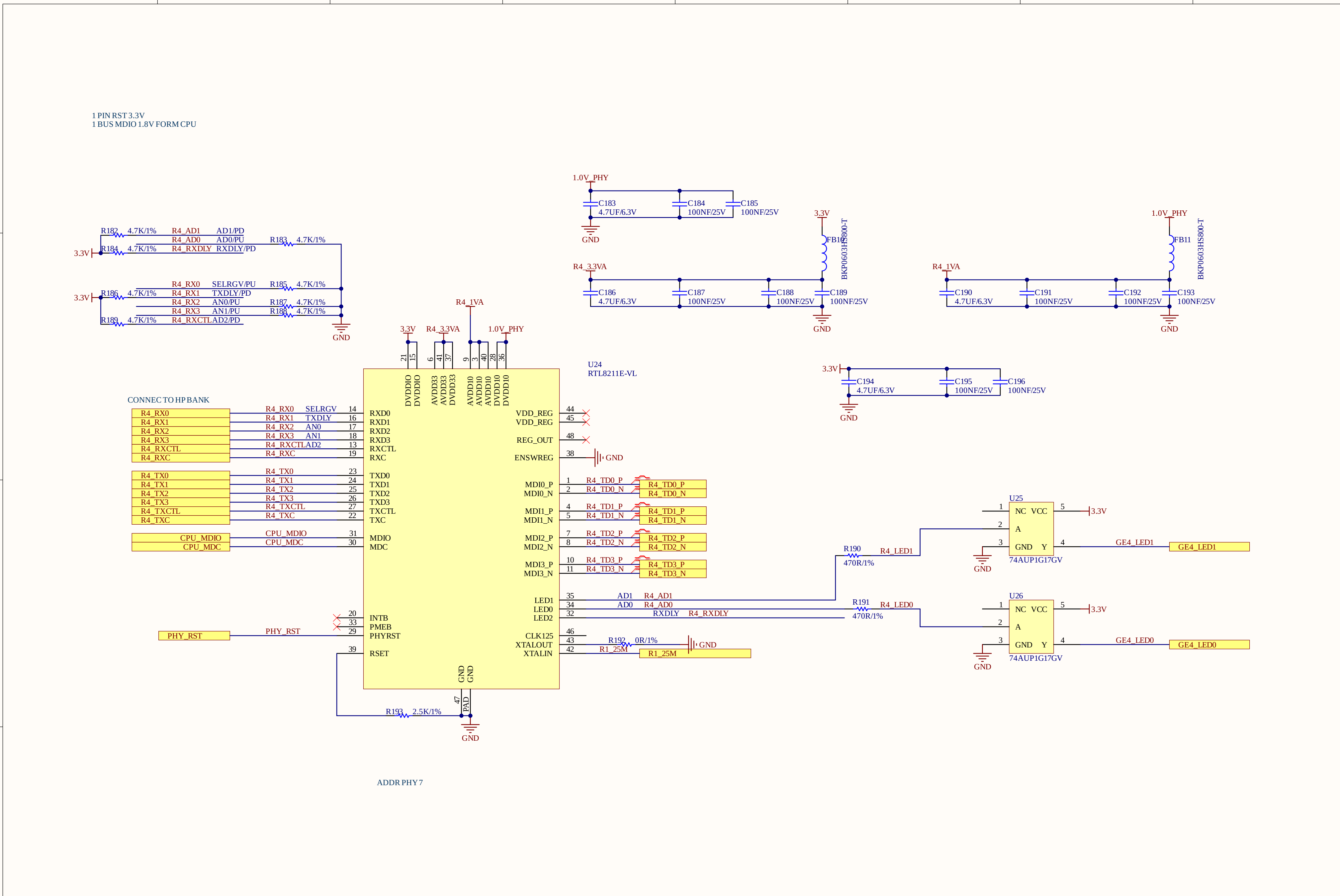
74 Pham Van Dong, Bac Tu Liem
Hanoi



1 PIN RST 3.3V
1 BUS MDIO 1.8V FORM CPU
R170 4.7K/1% R3_AD1 AD1/PD
R3_AD0 AD0/PU R171 4.7K/1%
3.3V R172 4.7K/1% R3_RXDLY RXDLY/PD
R3_RX0 SELRGV/PU R173 4.7K/1%
3.3V R174 4.7K/1% R3_RX1 TXDLY/PD
R3_RX2 AN0/PU R175 4.7K/1%
R3_RX3 AN1/PU R176 4.7K/1%
R177 4.7K/1% R3_RXCTL AD2/PD
GND
1.0V_PHY C169 4.7UF/6.3V C170 100NF/25V C171 100NF/25V GND
3.3V BKP0603HS800-T FB8
R3_3.3VA C172 4.7UF/6.3V C173 100NF/25V C174 100NF/25V C175 100NF/25V GND
1.0V_PHY BKP0603HS800-T FB9
R3_1VA C176 4.7UF/6.3V C177 100NF/25V C178 100NF/25V C179 100NF/25V GND
3.3V C180 4.7UF/6.3V C181 100NF/25V C182 100NF/25V GND
R3_1VA 3.3V R3_3.3VA 1.0V_PHY
U21
RTL8211E-VL
DVDDIO DVDDIO AVDD33 AVDD33 AVDD33 AVDD10 AVDD10 AVDD10 DVDD10 DVDD10
CONNECT TO HP BANK
R3_RX0 R3_RX1 R3_RX2 R3_RX3 R3_RXCTL R3_RXC
R3_RX0 SELRGV 14 RXD0
R3_RX1 TXDLY 16 RXD1
R3_RX2 AN0 17 RXD2
R3_RX3 AN1 18 RXD3
R3_RXCTL AD2 13 RXCTL
R3_RXC 19 RXC
R3_TX0 R3_TX1 R3_TX2 R3_TX3 R3_TXCTL R3_TXC
R3_TX0 23 TXD0
R3_TX1 24 TXD1
R3_TX2 25 TXD2
R3_TX3 26 TXD3
R3_TXCTL 27 TXCTL
R3_TXC 22 TXC
CPU_MDIO CPU_MDC
CPU_MDIO 31 MDIO
CPU_MDC 30 MDC
20 INTB
33 PMEB
PHY_RST PHY_RST 29 PHYRST
39 RSET
R181 2.5K/1%
GND GND 47 PAD GND
VDD_REG 44
VDD_REG 45
REG_OUT 48
ENSWREG 38 GND
MDI0_P 1 R3_TD0_P R3_TD0_P
MDI0_N 2 R3_TD0_N R3_TD0_N
MDI1_P 4 R3_TD1_P R3_TD1_P
MDI1_N 5 R3_TD1_N R3_TD1_N
MDI2_P 7 R3_TD2_P R3_TD2_P
MDI2_N 8 R3_TD2_N R3_TD2_N
MDI3_P 10 R3_TD3_P R3_TD3_P
MDI3_N 11 R3_TD3_N R3_TD3_N
LED1 35 AD1 R3_AD1
LED0 34 AD0 R3_AD0
LED2 32 RXDLY R3_RXDLY
CLK125 46
XTALOUT 43 R180 0R/1% GND
XTALIN 42 R2_25M R2_25M
R178 470R/1% R3_LED1
U22 NC VCC A GND Y 74AUP1G17GV 3.3V GE3_LED1 GE3_LED1 GND
R179 470R/1% R3_LED0
U23 NC VCC A GND Y 74AUP1G17GV 3.3V GE3_LED0 GE3_LED0 GND
ADDR PHY 7
Cannot open file C:\Users\esys\Dropbox\kinte
project: HTPSR5G
Size: A4 Number: 20 Revision: 1.0
Date: 9/30/2024 Time: 5:16:02 PM Sheet 20 of 41
File: PHY3.SchDoc
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



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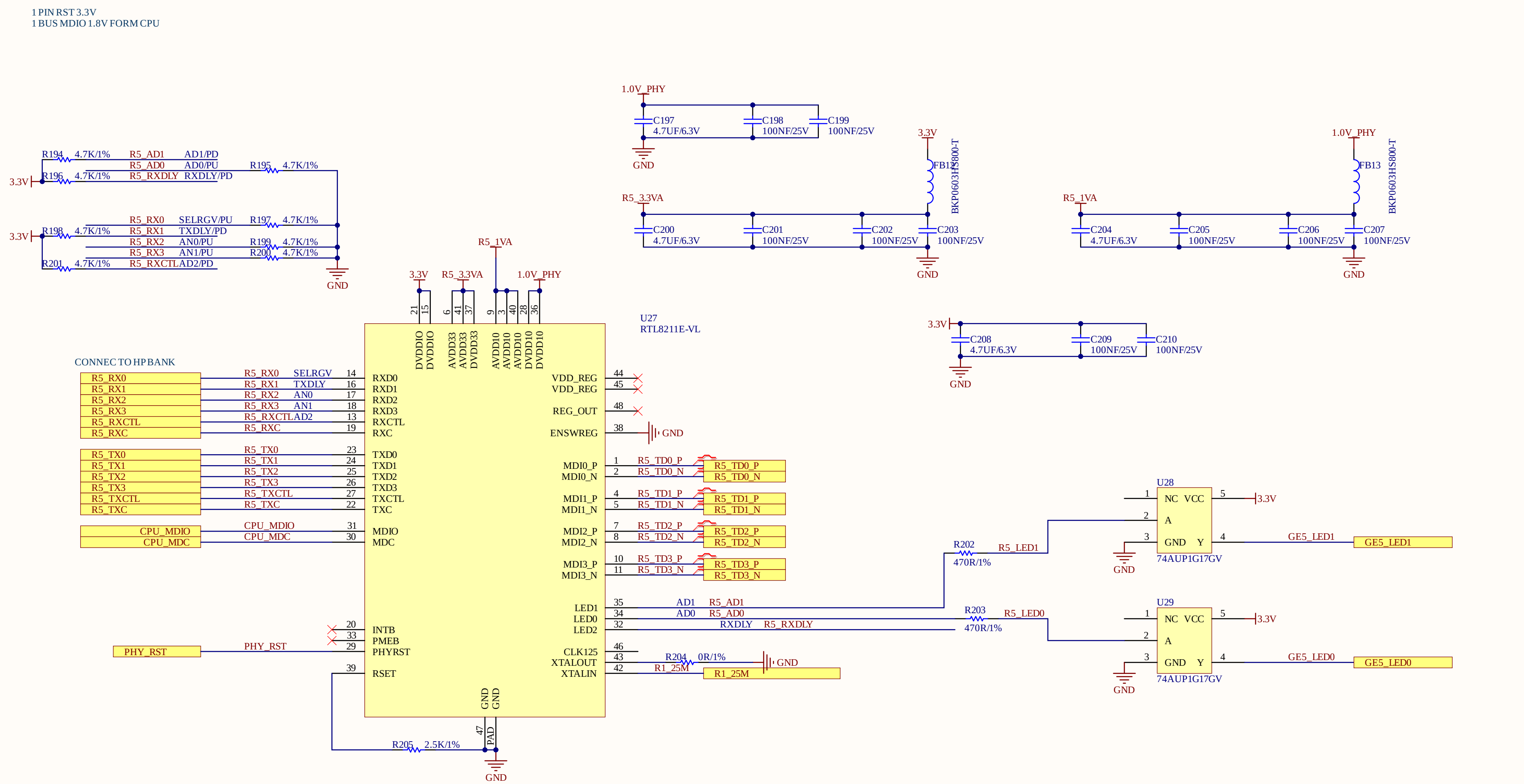
	Size: A4	Number: 21	Revision: 1.0
	Date: 9/30/2024	Time: 5:16:02 PM	Sheet 21 of 41
	File: PHY4.SchDoc		*OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi*



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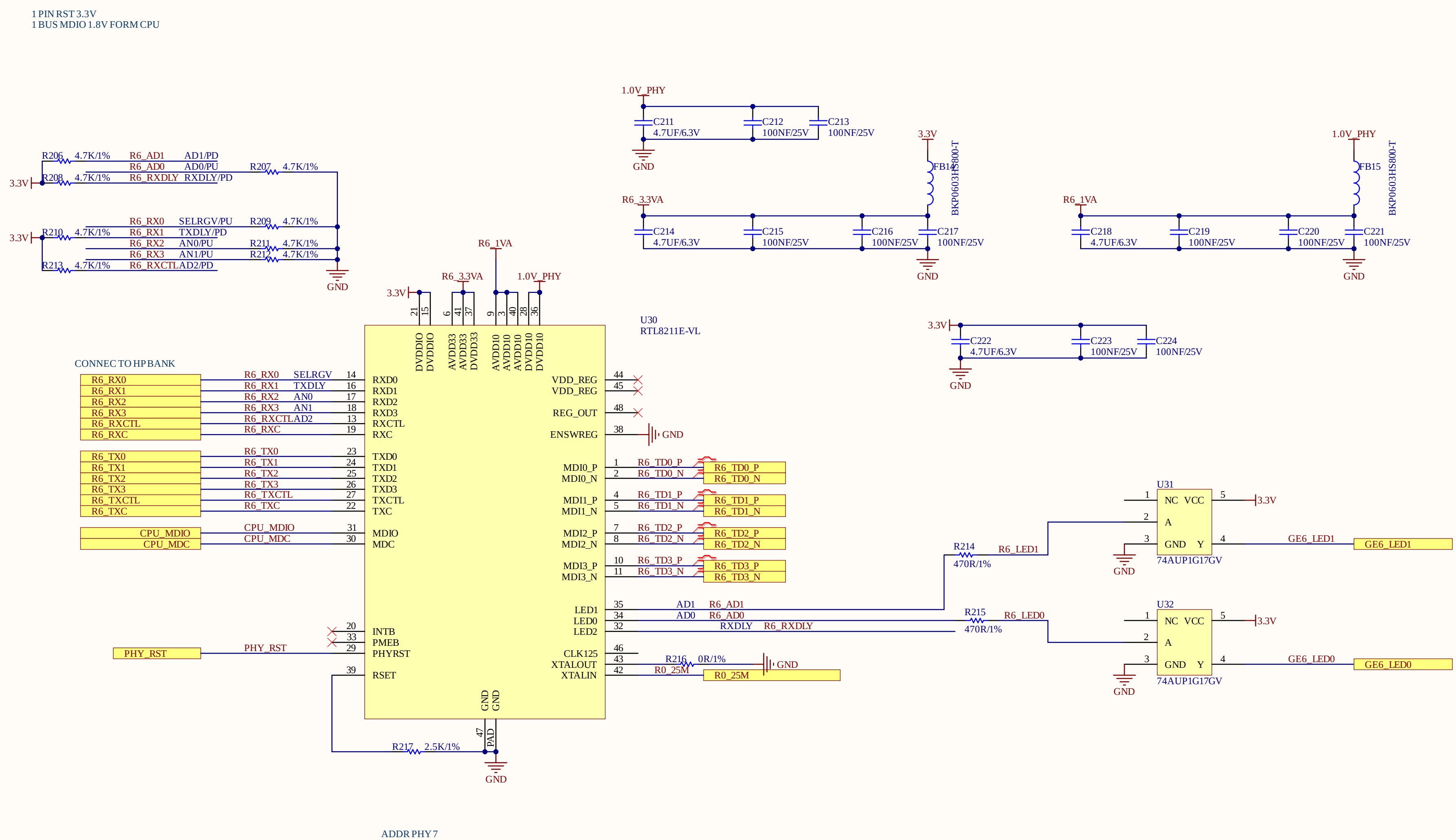
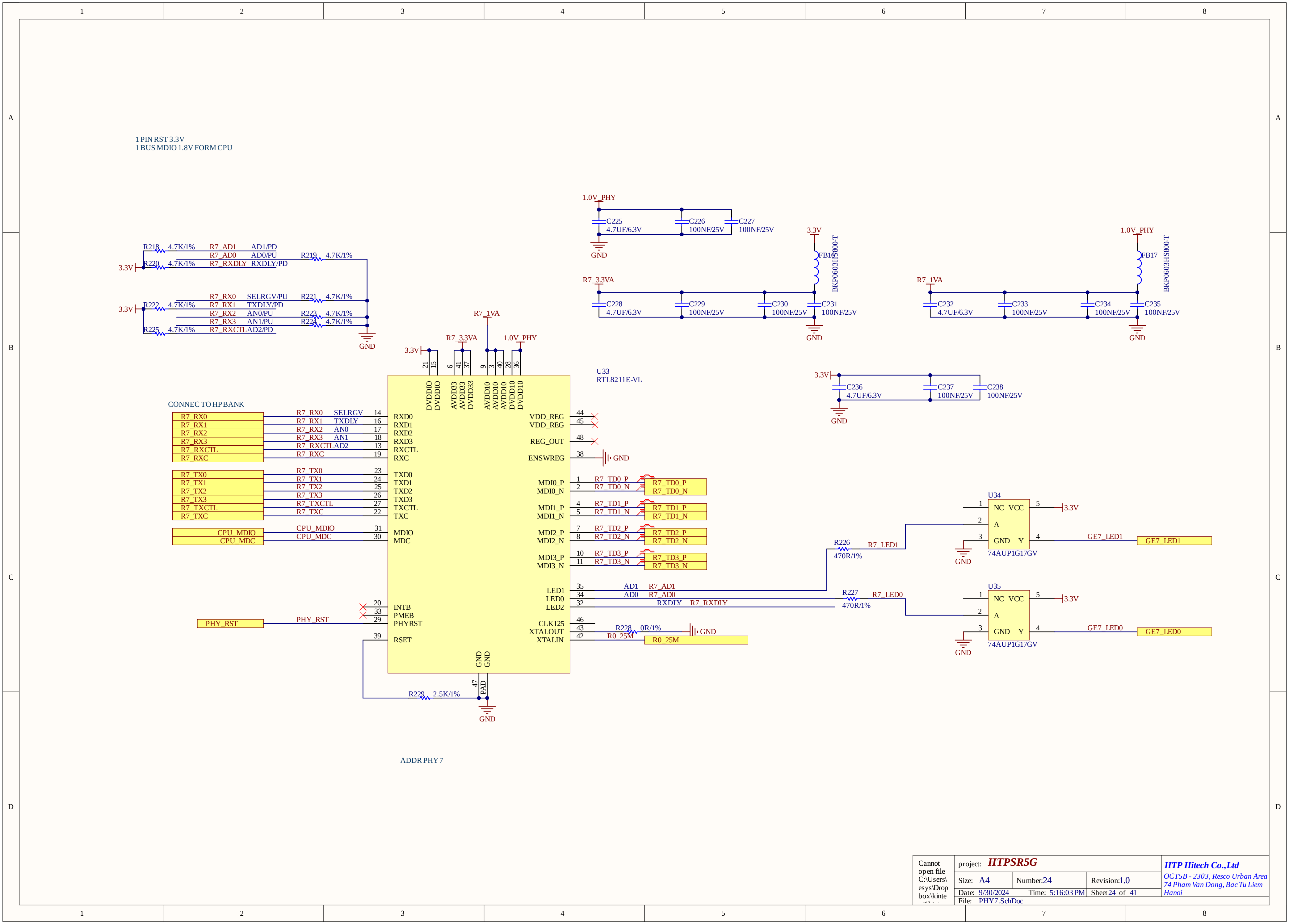
	Size: A4 Number:22 Revision:*	OCT5B - 2303, Resco Urban Area
	Date: 9/30/2024 Time: 5:16:03 PM Sheet 22 of 41	74 Pham Van Dong, Bac Tu Liem
	File: PHY5.SchDoc	Hanoi



project: HTPSR5G			HTP Hitech Co.,Ltd
Size: A4	Number: 23	Revision: 1.0	OCT5B - 2303, Resco Urban Area
Date: 9/30/2024	Time: 5:16:03 PM	Sheet 23 of 41	74 Pham Van Dong, Bac Tu Liem Hanoi
File: PHY6.SchDoc			



1 PIN RST 3.3V
1 BUS MDIO 1.8V FORM CPU
R218 4.7K/1% R7_AD1 AD1/PD
R7_AD0 AD0/PU R219 4.7K/1%
R220 4.7K/1% R7_RXDLY RXDLY/PD
R7_RX0 SELRGV/PU R221 4.7K/1%
R222 4.7K/1% R7_RX1 TXDLY/PD
R7_RX2 AN0/PU R223 4.7K/1%
R7_RX3 AN1/PU R224 4.7K/1%
R225 4.7K/1% R7_RXCTL AD2/PD
3.3V
GND
1.0V_PHY
C225 4.7UF/6.3V
C226 100NF/25V
C227 100NF/25V
R7_3.3VA
C228 4.7UF/6.3V
C229 100NF/25V
C230 100NF/25V
C231 100NF/25V
FB16 BKP0603HS800-T
R7_1VA
C232 4.7UF/6.3V
C233 100NF/25V
C234 100NF/25V
C235 100NF/25V
FB17 BKP0603HS800-T
C236 4.7UF/6.3V
C237 100NF/25V
C238 100NF/25V
U33
RTL8211E-VL
DVDDIO DVDDIO AVDD33 AVDD33 AVDD33 AVDD10 AVDD10 AVDD10 DVDD10 DVDD10
CONNEC TO HP BANK
R7_RX0 R7_RX1 R7_RX2 R7_RX3 R7_RXCTL R7_RXC
R7_TX0 R7_TX1 R7_TX2 R7_TX3 R7_TXCTL R7_TXC
CPU_MDIO CPU_MDC
PHY_RST
RXD0 RXD1 RXD2 RXD3 RXCTL RXC
TXD0 TXD1 TXD2 TXD3 TXCTL TXC
MDIO MDC
INTB PMEB PHYRST RSET
VDD_REG VDD_REG REG_OUT ENSWREG
MDI0_P MDI0_N MDI1_P MDI1_N MDI2_P MDI2_N MDI3_P MDI3_N
R7_TD0_P R7_TD0_N R7_TD1_P R7_TD1_N R7_TD2_P R7_TD2_N R7_TD3_P R7_TD3_N
LED1 LED0 LED2
CLK125 XTALOUT XTALIN
R228 0R/1%
R0_25M
R229 2.5K/1%
PAD
U34
NC VCC A GND Y
74AUP1G17GV
GE7_LED1
R226 470R/1% R7_LED1
U35
74AUP1G17GV
GE7_LED0
R227 470R/1% R7_LED0
ADDR PHY 7
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project: HTPSR5G
Size: A4
Number: 24
Revision: 1.0
Date: 9/30/2024
Time: 5:16:03 PM
Sheet 24 of 41
File: PHY7.SchDoc
HTP Hitech Co.,Ltd
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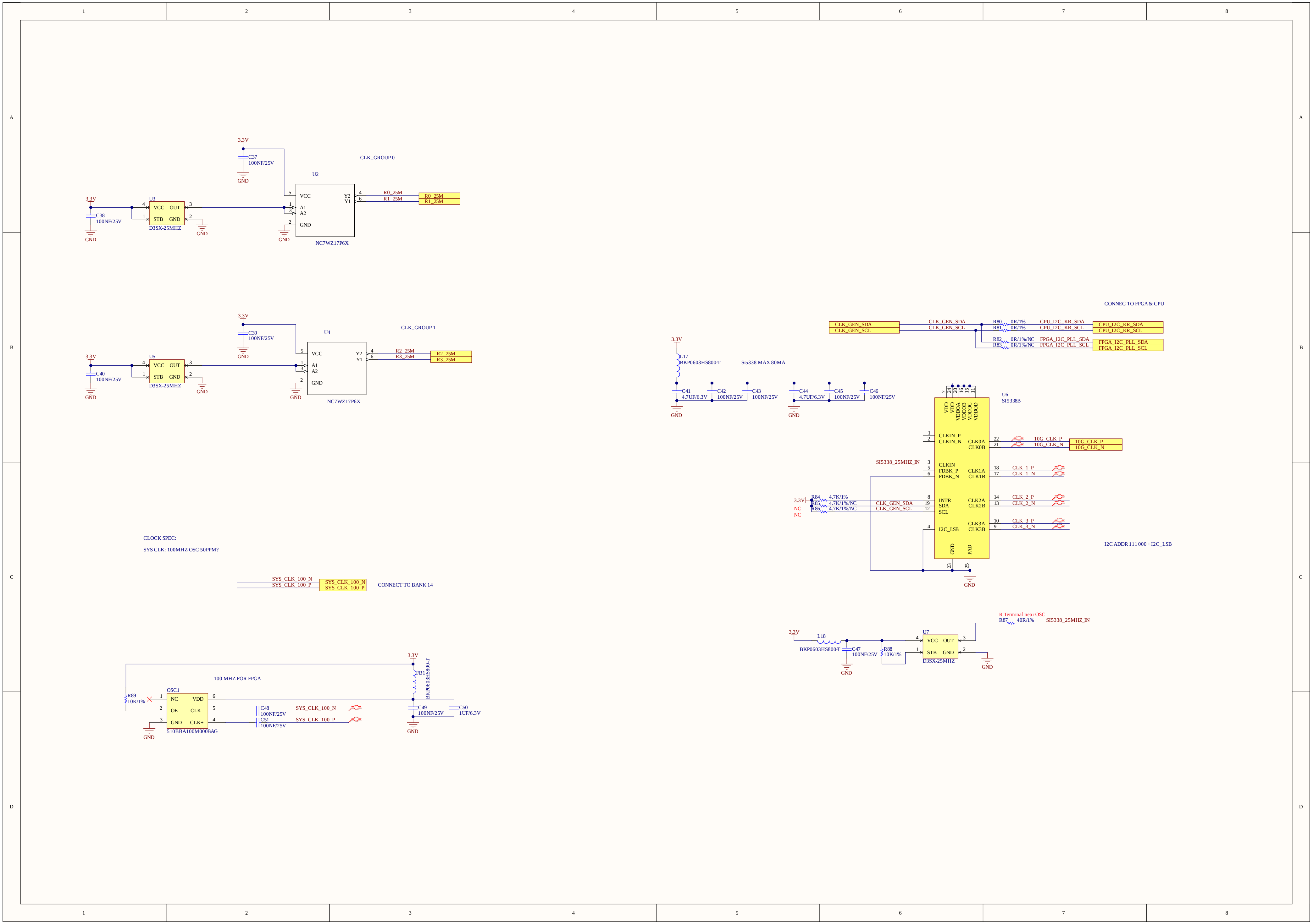
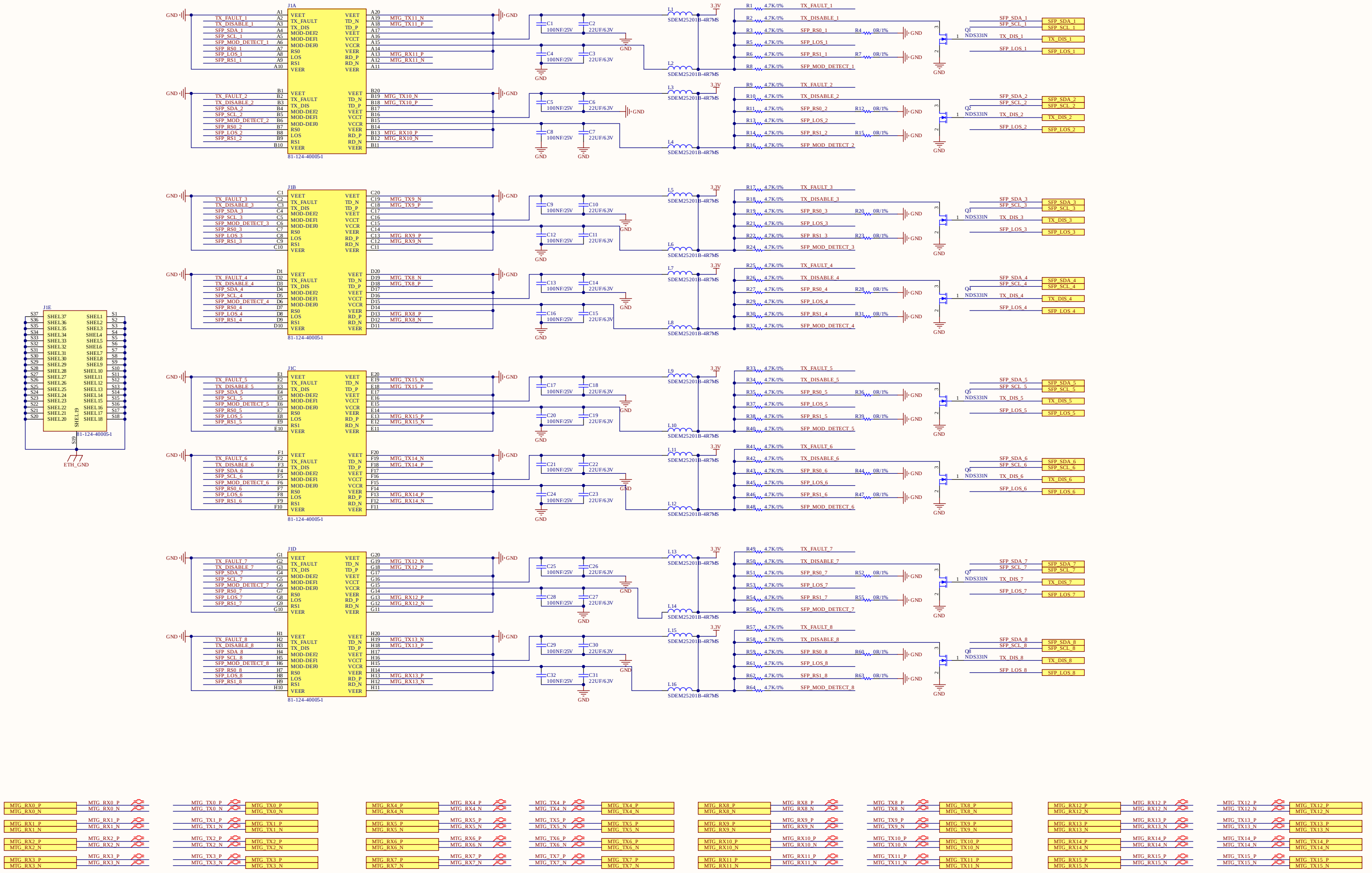
74 Pham Van Dong, Bac Tu Liem
Hanoi



CLK_GROUP 0
U2
U3
VCC OUT
STB GND
D3SX-25MHZ
C37 100NF/25V
C38 100NF/25V
3.3V
GND
VCC
A1
A2
GND
Y2
Y1
R0_25M
R1_25M
NC7WZ17P6X
CLK_GROUP 1
U4
U5
C39 100NF/25V
C40 100NF/25V
R2_25M
R3_25M
NC7WZ17P6X
CONNEC TO FPGA & CPU
CLK_GEN_SDA
CLK_GEN_SCL
R80 0R/1%
R81 0R/1%
R82 0R/1%/NC
R83 0R/1%/NC
CPU_I2C_KR_SDA
CPU_I2C_KR_SCL
FPGA_I2C_PLL_SDA
FPGA_I2C_PLL_SCL
L17
BKP0603HS800-T
SI5338 MAX 80MA
C41 4.7UF/6.3V
C42 100NF/25V
C43 100NF/25V
C44 4.7UF/6.3V
C45 100NF/25V
C46 100NF/25V
U6
SI5338B
VDD VDD VDDOA VDDOB VDDOC VDDOD
CLKIN_P
CLKIN_N
CLK0A
CLK0B
10G_CLK_P
10G_CLK_N
SI5338_25MHZ_IN
CLKIN
FDBK_P
FDBK_N
CLK1A
CLK1B
CLK_1_P
CLK_1_N
R84 4.7K/1%
R85 4.7K/1%/NC
R86 4.7K/1%/NC
NC
INTR
SDA
SCL
CLK2A
CLK2B
CLK_2_P
CLK_2_N
CLK3A
CLK3B
CLK_3_P
CLK_3_N
I2C_LSB
GND
PAD
I2C ADDR 111 000 +I2C_LSB
CLOCK SPEC:
SYS CLK: 100MHZ OSC 50PPM?
SYS_CLK_100_N
SYS_CLK_100_P
CONNECT TO BANK 14
R Terminal near OSC
R87 40R/1%
L18
BKP0603HS800-T
C47 100NF/25V
R88 10K/1%
U7
D3SX-25MHZ
100 MHZ FOR FPGA
R89 10K/1%
OSC1
NC VDD
OE CLK-
GND CLK+
510BBA100M000BAG
C48 100NF/25V
C51 100NF/25V
FB1
BKP0603HS800-T
C49 100NF/25V
C50 1UF/6.3V



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	Size: A4	Number: 27	Revision: 1.0
	Date: 9/30/2024 Time: 5:16:04 PM	Sheet27 of 41	OCT5B-2303,Resco Urban Area 74 Pham Van Dong,Bac Tu Liem Hanoi
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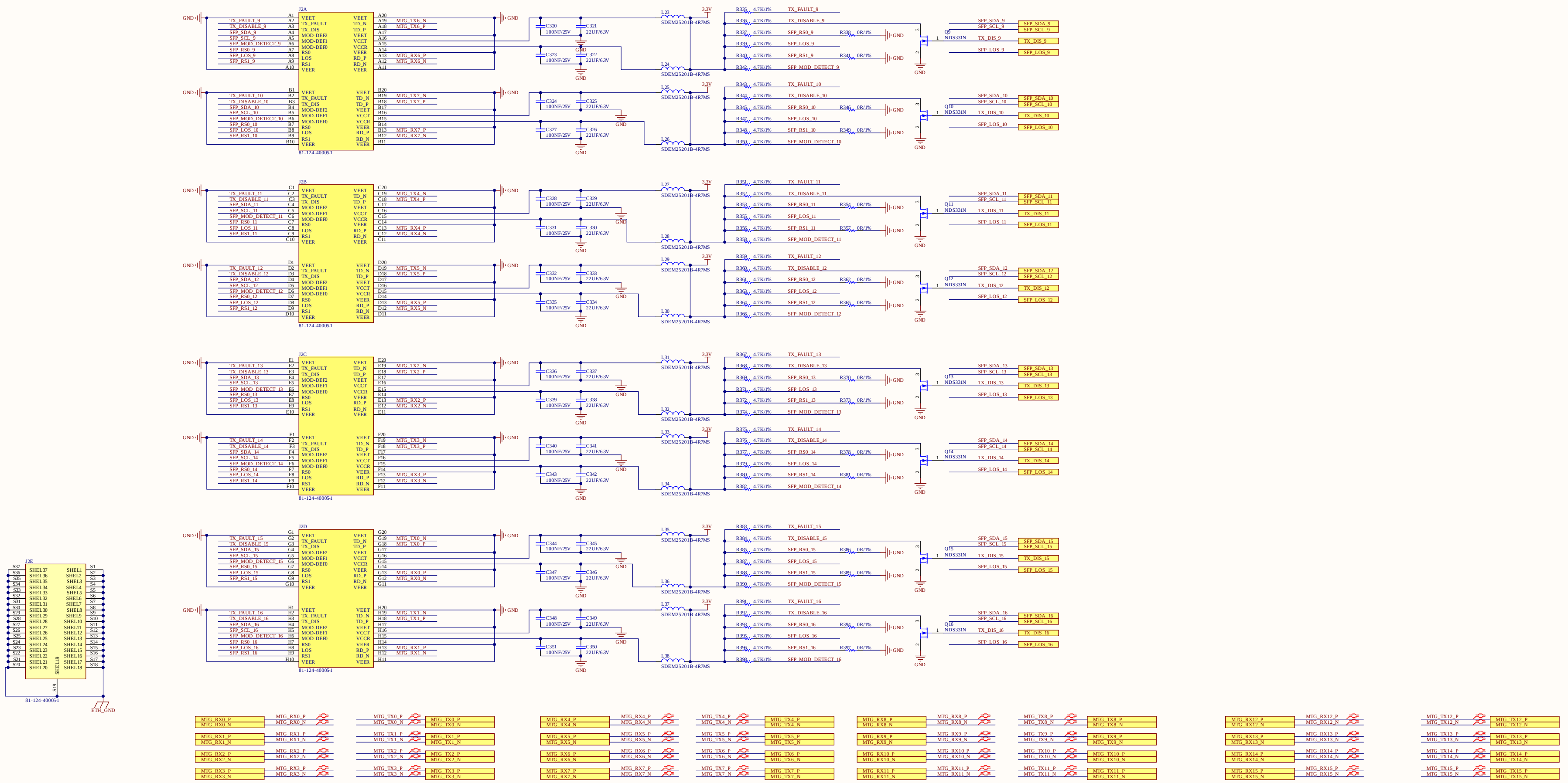

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	Size: A4	Number: 28	Revision: 1.0	OCT5B-2303,Resco Urban Area 74 Pham Van Dong,Bac Tu Liem Hanoi
	Date: 9/30/2024 Time: 5:16:05 PM		Sheet 28 of 41	
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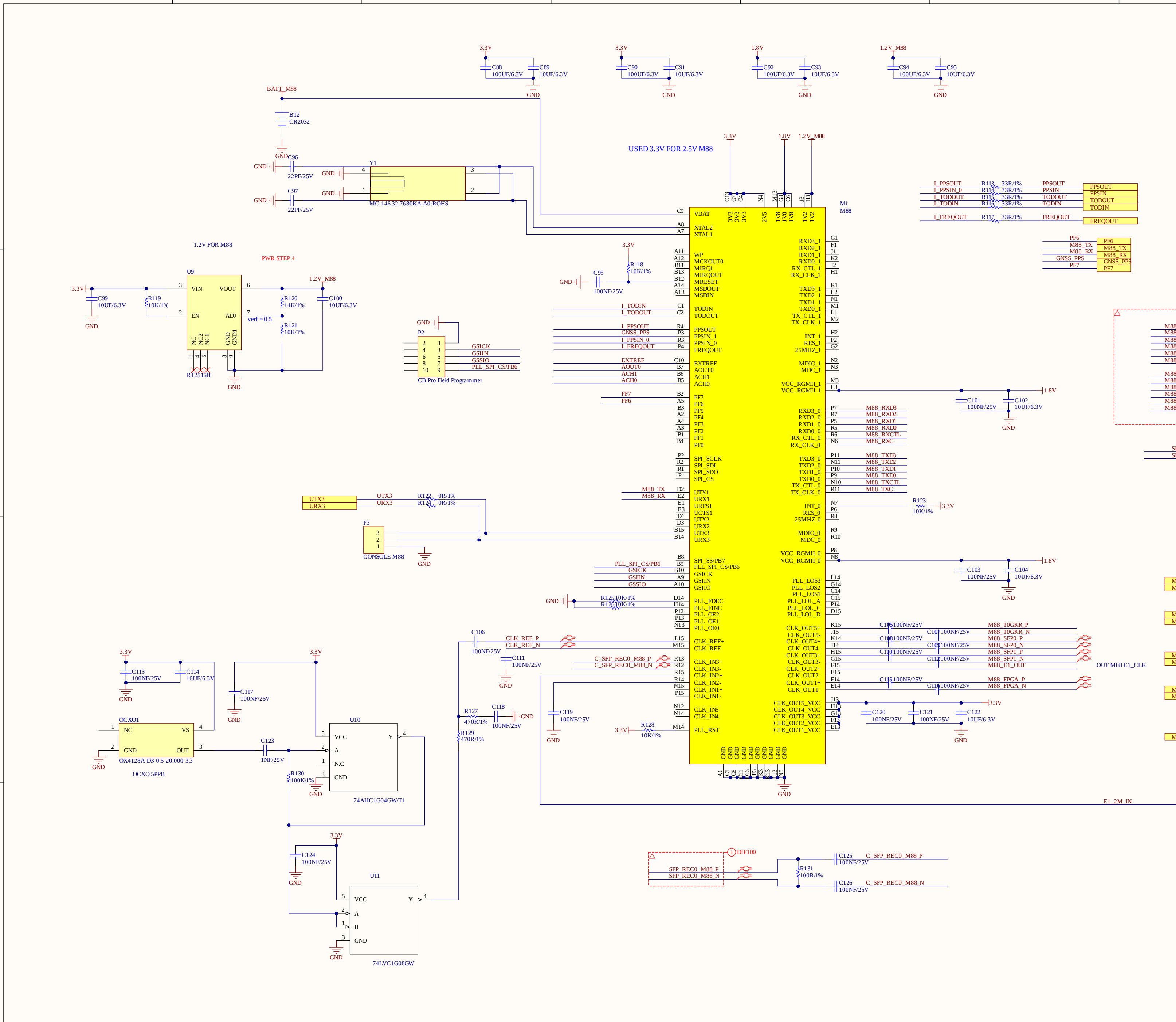
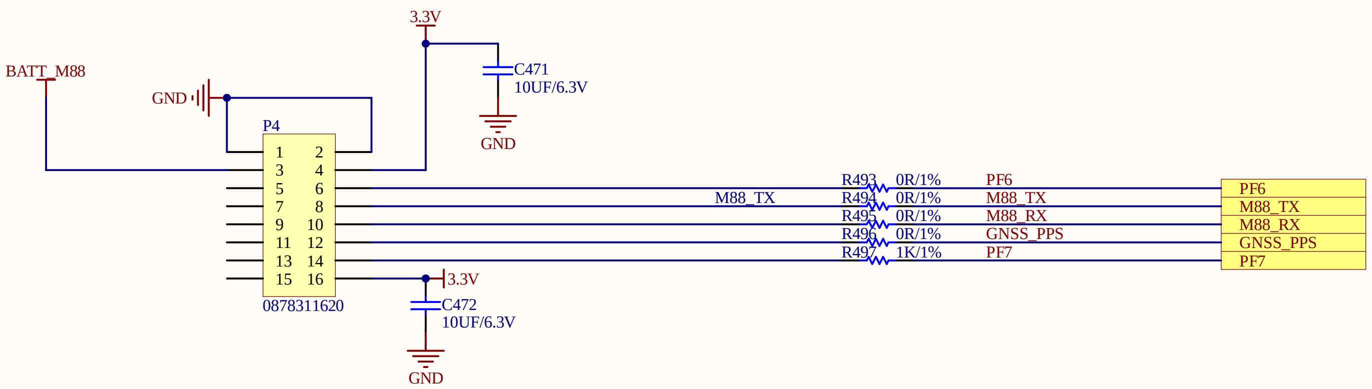


project: HTPSR5G
Size: A4 Number: 29 Revision: 1.0
Date: 9/30/2024 Time: 5:16:05 PM Sheet 29 of 41
File: M88 MODULE.SchDoc

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



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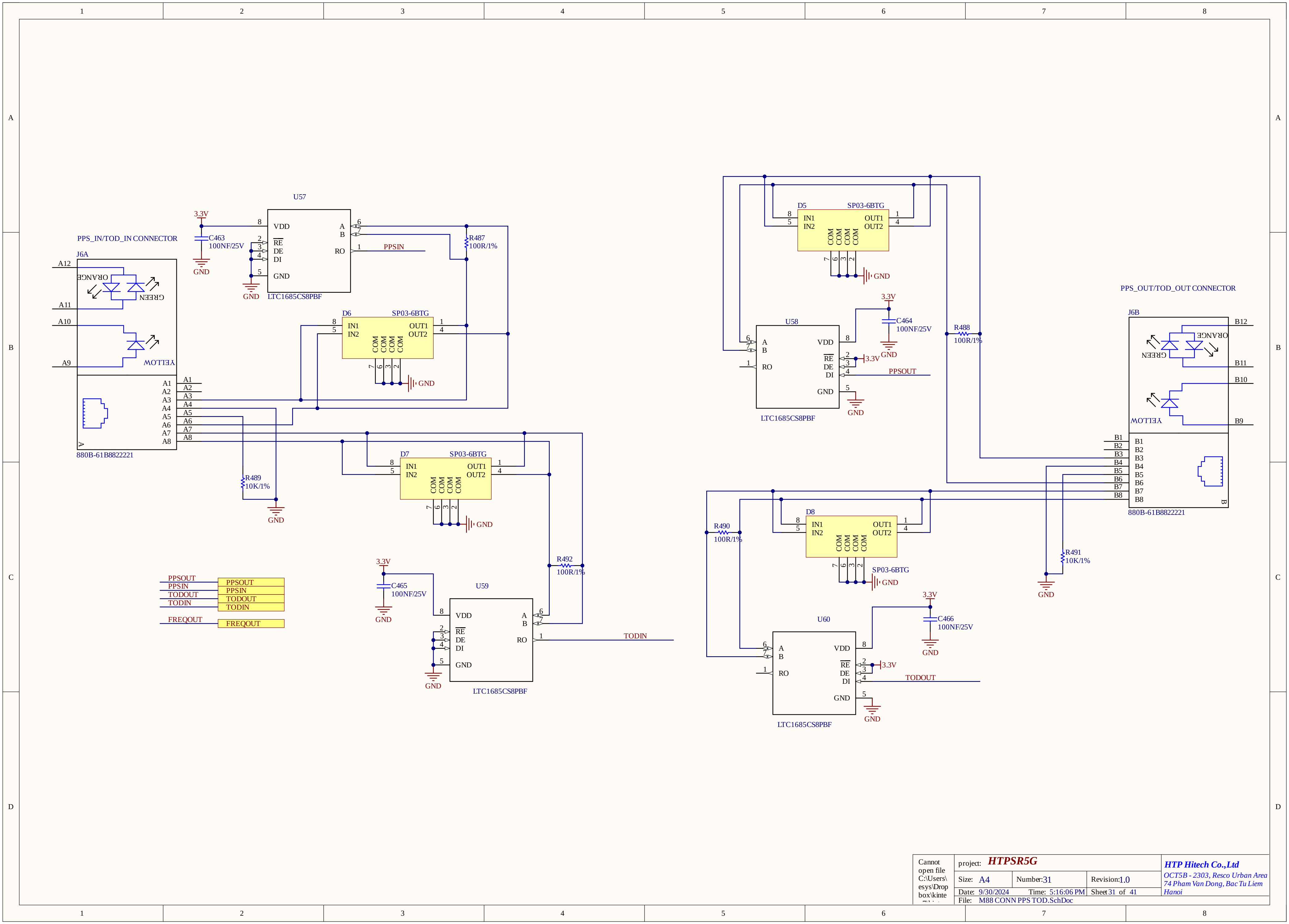
project: ***HTPSR5G***

Size: A4 | Number: 30 | Revision: 1.0

Date: 9/30/2024 | Time: 5:16:05 PM | Sheet 30 of 41

File: M88 GNSS.SchDoc

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi


PPS_IN/TOD_IN CONNECTOR
J6A
ORANGE
GREEN
YELLOW
880B-61B8822221
U57
LTC1685CS8PBF
C463
100NF/25V
3.3V
GND
PPSIN
R487
100R/1%
D6
SP03-6BTG
R489
10K/1%
D7
SP03-6BTG
C465
100NF/25V
U59
LTC1685CS8PBF
R492
100R/1%
TODIN
PPSOUT
PPSIN
TODOUT
TODIN
FREQOUT
D5
SP03-6BTG
U58
LTC1685CS8PBF
C464
100NF/25V
R488
100R/1%
PPSOUT
PPS_OUT/TOD_OUT CONNECTOR
J6B
880B-61B8822221
R490
100R/1%
D8
SP03-6BTG
R491
10K/1%
C466
100NF/25V
U60
LTC1685CS8PBF
TODOUT
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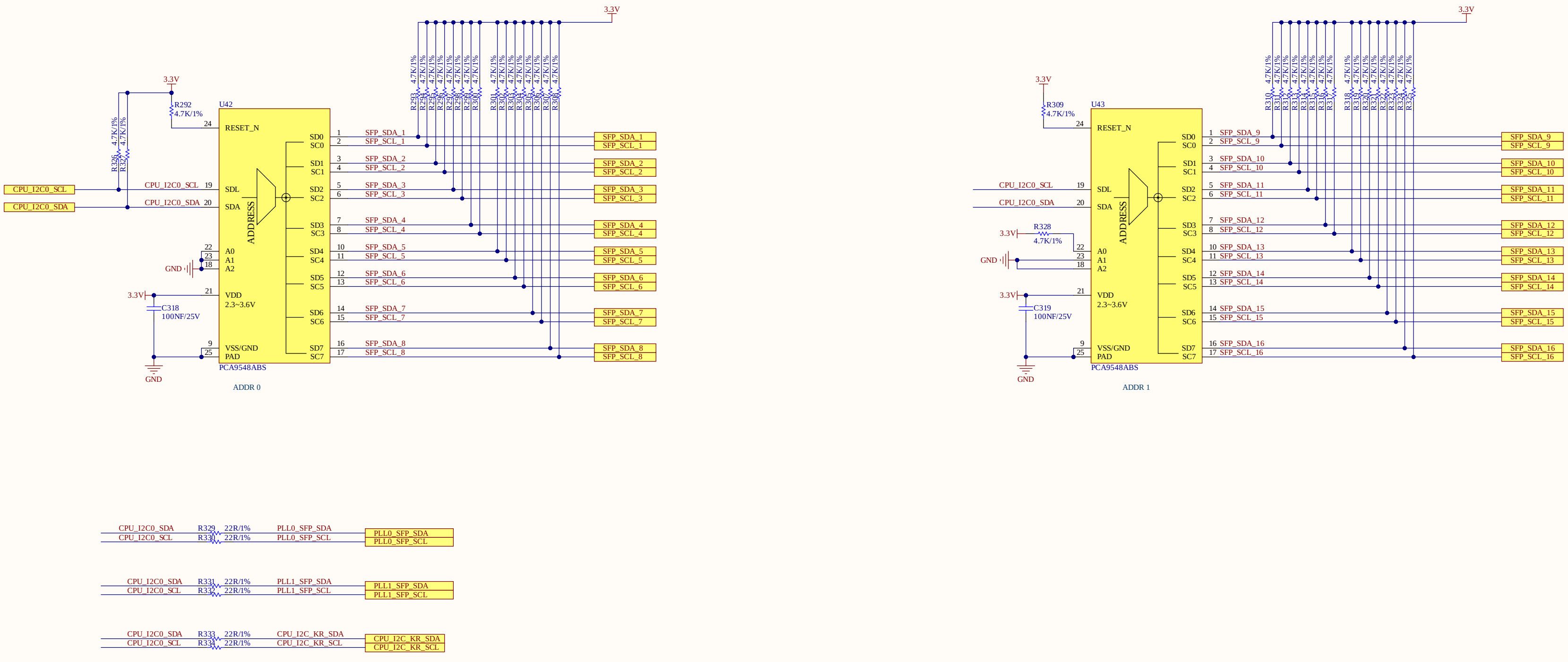
project: HTPSR5G
Size: A4
Number: 31
Revision: 1.0
Date: 9/30/2024
Time: 5:16:06 PM
Sheet 31 of 41
File: M88 CONN PPS TOD.SchDoc
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



project: **HTPSR5G**			HTP Hitech Co.,Ltd
Size: A4	Number: 33	Revision: 1.0	OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi
Date: 9/30/2024	Time: 5:16:06 PM	Sheet 33 of 41	
File: SW I2C.SchDoc			

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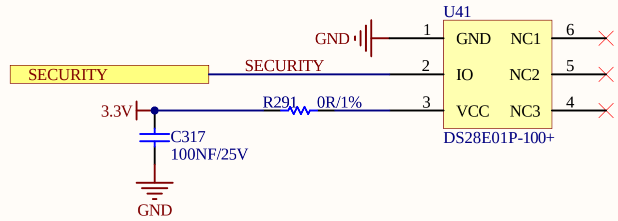

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	Size: A4	Number: 34	Revision: 1.0	*OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi*
	Date: 9/30/2024	Time: 5:16:06 PM	Sheet 34 of 41	
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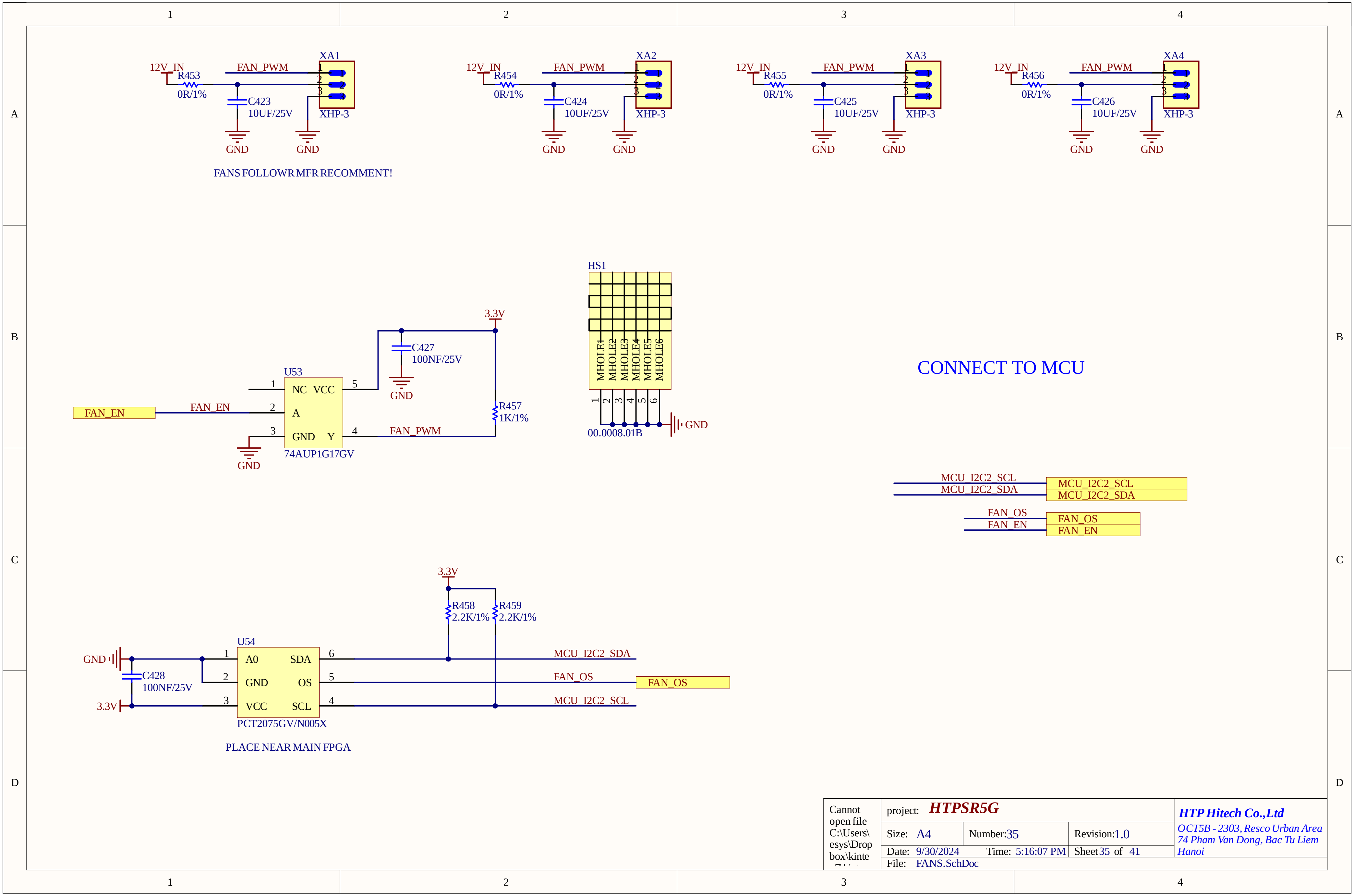
12V_IN
R453
0R/1%
FAN_PWM
XA1
C423
10UF/25V
XHP-3
GND
GND
12V_IN
R454
0R/1%
FAN_PWM
XA2
C424
10UF/25V
XHP-3
GND
GND
12V_IN
R455
0R/1%
FAN_PWM
XA3
C425
10UF/25V
XHP-3
GND
GND
12V_IN
R456
0R/1%
FAN_PWM
XA4
C426
10UF/25V
XHP-3
GND
GND
FANS FOLLOWR MFR RECOMMENT!
HS1
MHOLE1
MHOLE2
MHOLE3
MHOLE4
MHOLE5
MHOLE6
00.0008.01B
GND
3.3V
C427
100NF/25V
GND
U53
NC VCC
A
GND Y
74AUP1G17GV
FAN_EN
FAN_EN
FAN_PWM
R457
1K/1%
GND
CONNECT TO MCU
MCU_I2C2_SCL
MCU_I2C2_SDA
MCU_I2C2_SCL
MCU_I2C2_SDA
FAN_OS
FAN_EN
FAN_OS
FAN_EN
3.3V
R458
2.2K/1%
R459
2.2K/1%
U54
GND
C428
100NF/25V
3.3V
A0 SDA
GND OS
VCC SCL
PCT2075GV/N005X
MCU_I2C2_SDA
FAN_OS
FAN_OS
MCU_I2C2_SCL
PLACE NEAR MAIN FPGA
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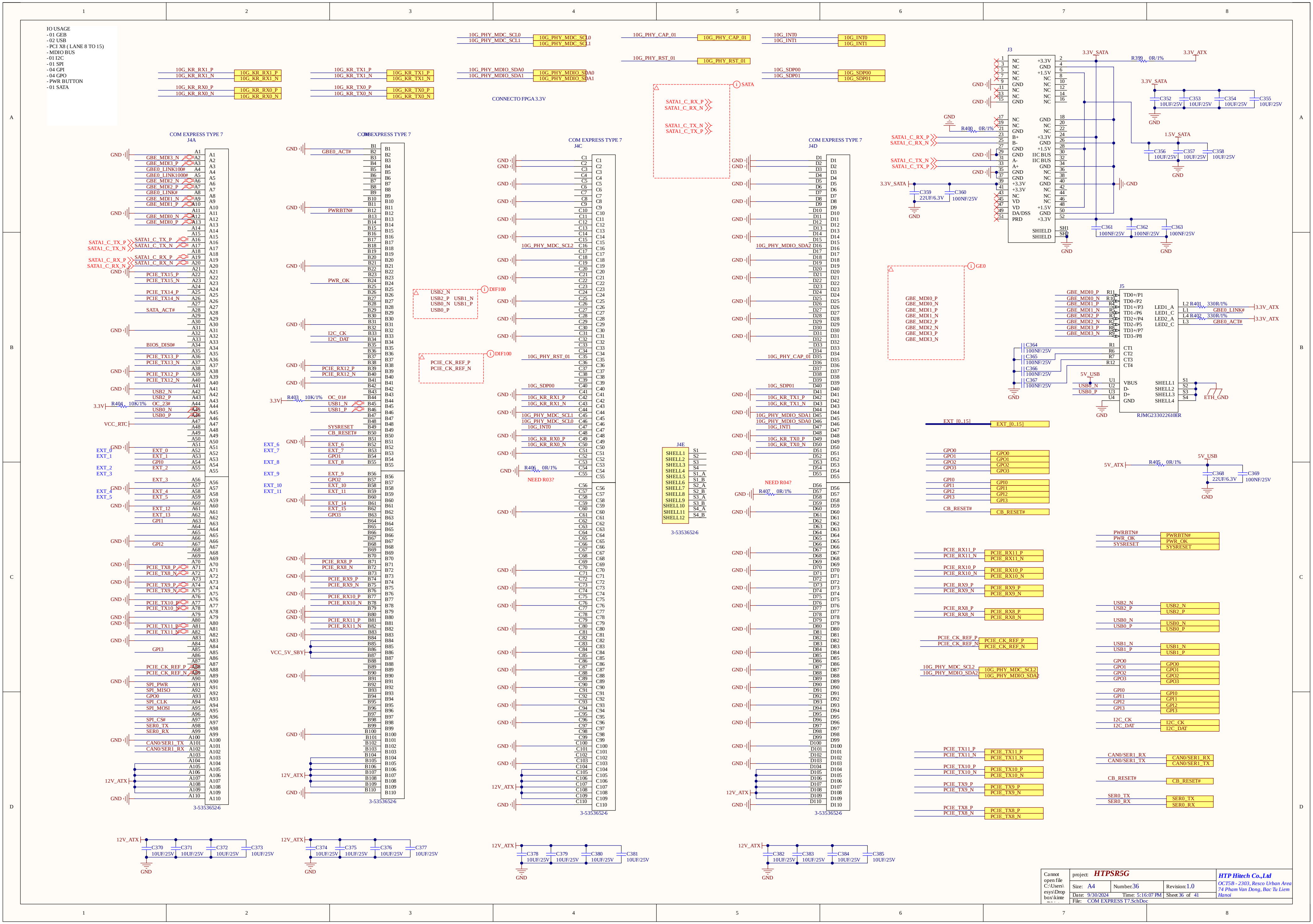
Size: A4
Number:35
Revision:1.0
Date: 9/30/2024
Time: 5:16:07 PM
Sheet35 of 41
File: FANS.SchDoc
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



IO USAGE
- 01 GEB
- 02 USB
- PCI X8 (LANE 8 TO 15)
- MDIO BUS
- 01 I2C
- 01 SPI
- 04 GPI
- 04 GPO
- PWR BUTTON
- 01 SATA
CONNECTO FPGA 3.3V
COM EXPRESS TYPE 7
J4A
J4C
J4D
J4E
3-5353652-6
J3
J5
RJMG233022610ER
NEED R03?
NEED R04?
DIF100
SATA
GE0
12V_ATX
3.3V_SATA
3.3V_ATX
1.5V_SATA
5V_ATX
5V_USB
VCC_5V_SBY
VCC_RTC
GND
ETH_GND
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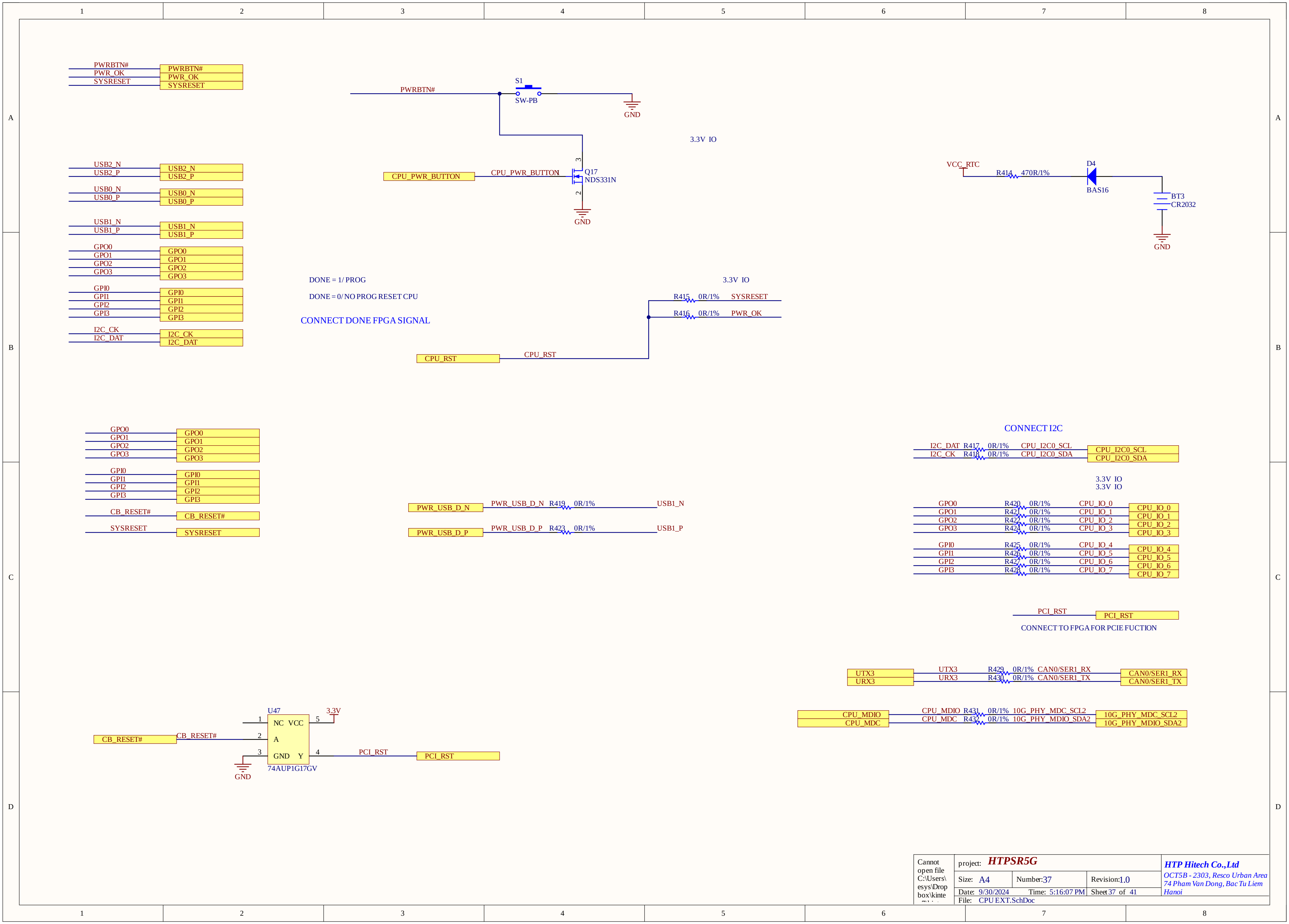
project: HTPSR5G
Size: A4
Number:36
Revision:1.0
Date: 9/30/2024
Time: 5:16:07 PM
Sheet 36 of 41
File: COM EXPRESS T7.SchDoc
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



PWRBTN#
PWR_OK
SYSRESET
PWRBTN#
S1
SW-PB
GND
3.3V IO
CPU_PWR_BUTTON
Q17
NDS331N
GND
VCC_RTC
R414 470R/1%
D4
BAS16
BT3
CR2032
GND
USB2_N
USB2_P
USB0_N
USB0_P
USB1_N
USB1_P
GPO0
GPO1
GPO2
GPO3
GPI0
GPI1
GPI2
GPI3
I2C_CK
I2C_DAT
DONE = 1/ PROG
DONE = 0/ NO PROG RESET CPU
CONNECT DONE FPGA SIGNAL
3.3V IO
R415 0R/1% SYSRESET
R416 0R/1% PWR_OK
CPU_RST
CONNECT I2C
I2C_DAT R417 0R/1% CPU_I2C0_SCL
I2C_CK R418 0R/1% CPU_I2C0_SDA
CPU_I2C0_SCL
CPU_I2C0_SDA
3.3V IO
3.3V IO
CB_RESET#
SYSRESET
PWR_USB_D_N
PWR_USB_D_N R419 0R/1% USB1_N
PWR_USB_D_P
PWR_USB_D_P R423 0R/1% USB1_P
GPO0 R420 0R/1% CPU_IO_0
GPO1 R421 0R/1% CPU_IO_1
GPO2 R422 0R/1% CPU_IO_2
GPO3 R424 0R/1% CPU_IO_3
GPI0 R425 0R/1% CPU_IO_4
GPI1 R426 0R/1% CPU_IO_5
GPI2 R427 0R/1% CPU_IO_6
GPI3 R428 0R/1% CPU_IO_7
PCI_RST
CONNECT TO FPGA FOR PCIE FUCTION
UTX3
URX3
UTX3 R429 0R/1% CAN0/SER1_RX
URX3 R430 0R/1% CAN0/SER1_TX
CAN0/SER1_RX
CAN0/SER1_TX
U47
3.3V
NC VCC
A
GND Y
74AUP1G17GV
CB_RESET#
PCI_RST
GND
CPU_MDIO
CPU_MDC
CPU_MDIO R431 0R/1% 10G_PHY_MDC_SCL2
CPU_MDC R432 0R/1% 10G_PHY_MDIO_SDA2
10G_PHY_MDC_SCL2
10G_PHY_MDIO_SDA2
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project: HTPSR5G
Size: A4
Number: 37
Revision: 1.0
Date: 9/30/2024
Time: 5:16:07 PM
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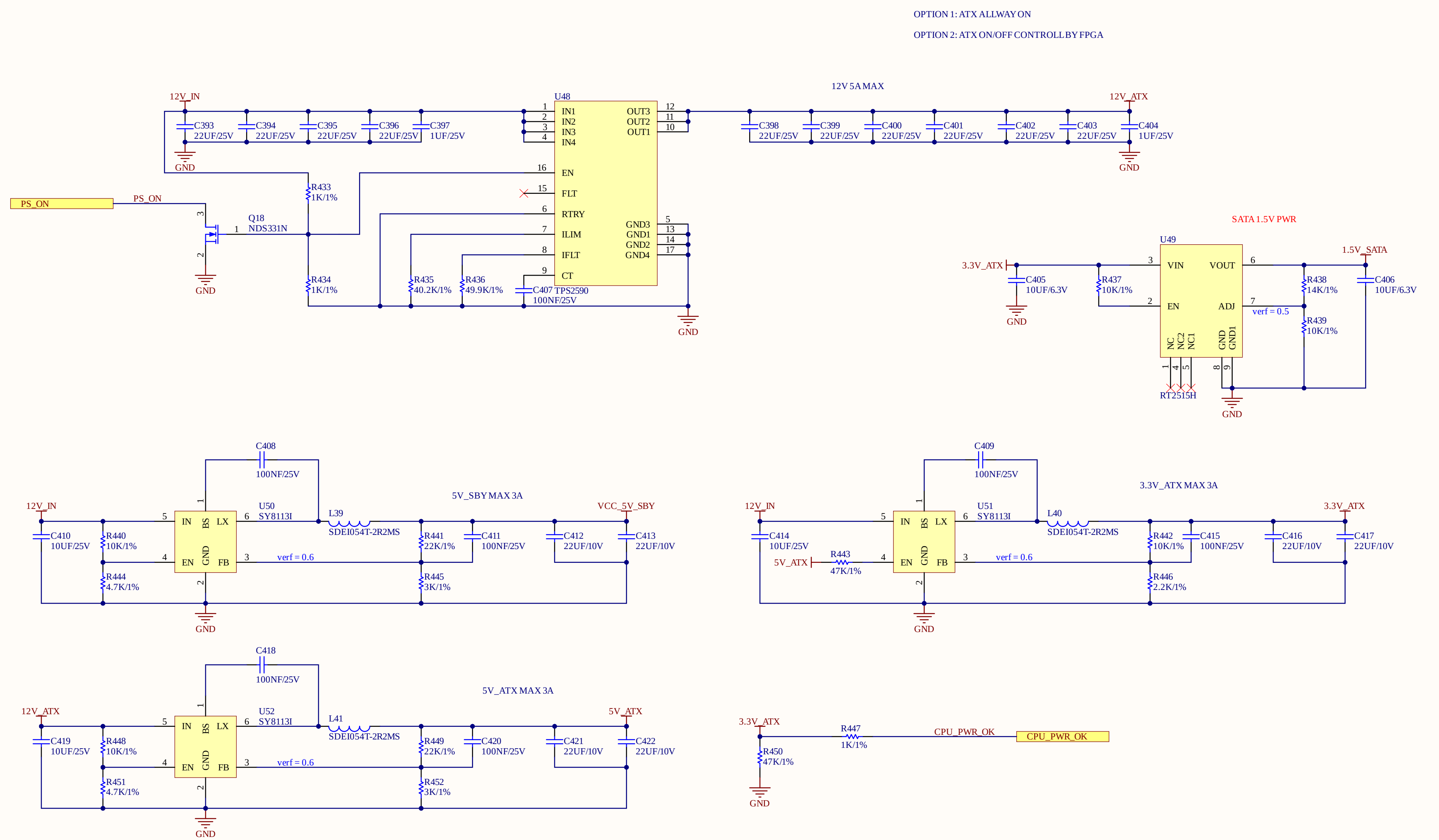
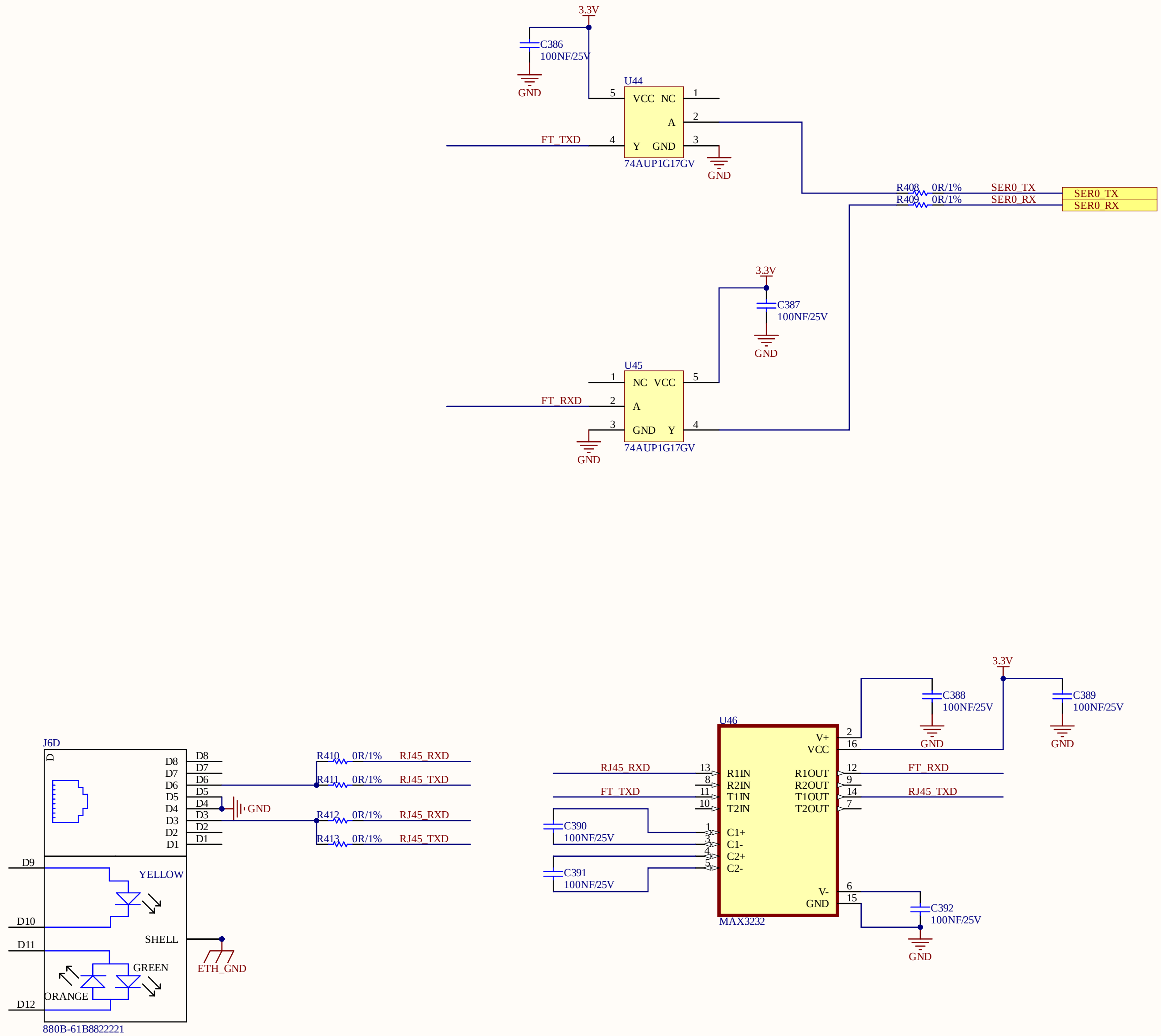
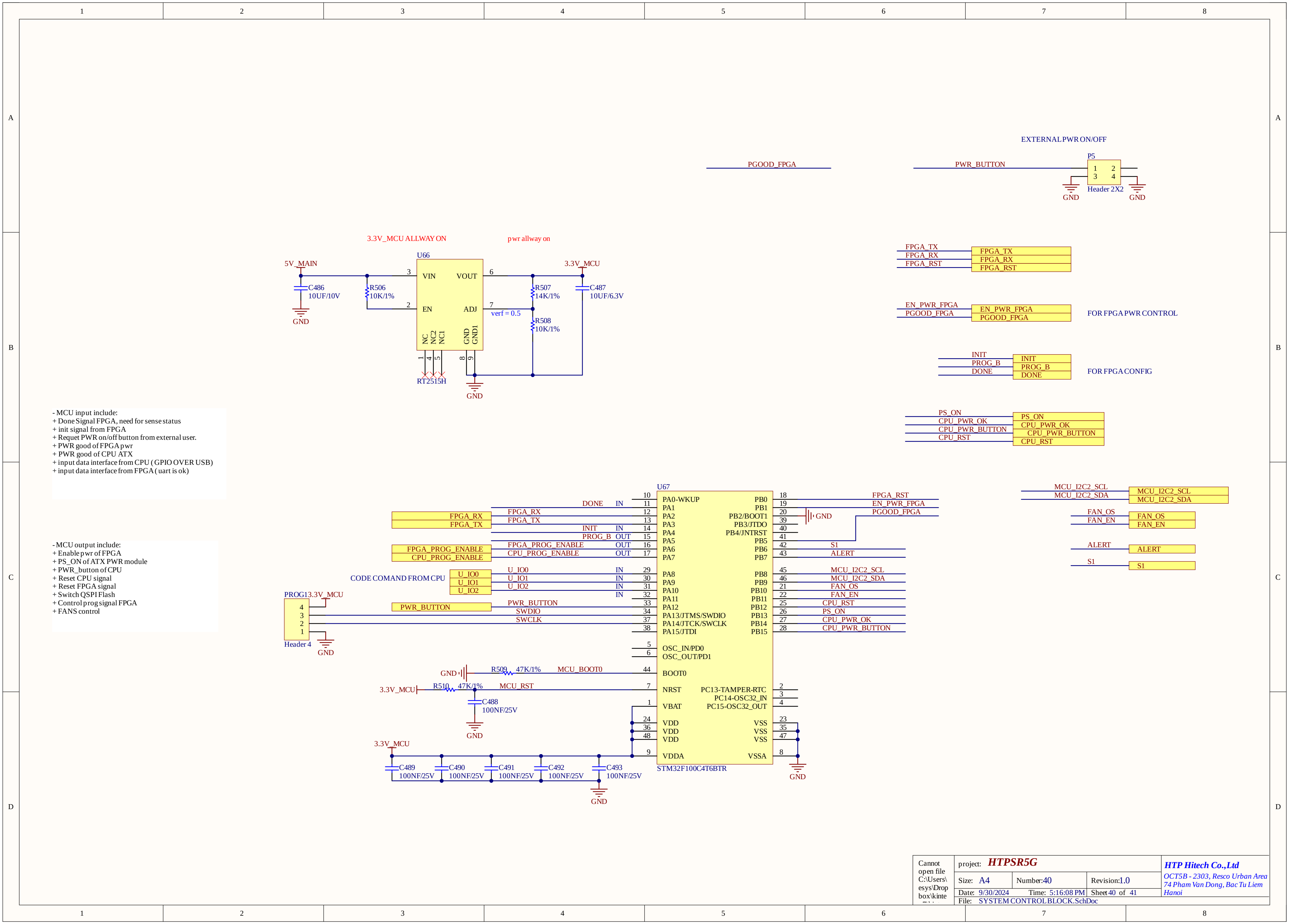
File: CPU EXT.SchDoc
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



Cannot open file C:\Users\esys\Drop box\kinte	project: HTPSR5G			HTP Hitech Co.,Ltd OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi
	Size: A4	Number: 38	Revision: 1.0	
	Date: 9/30/2024	Time: 5:16:08 PM	Sheet 38 of 41	
	File: CPU PWR.SchDoc			



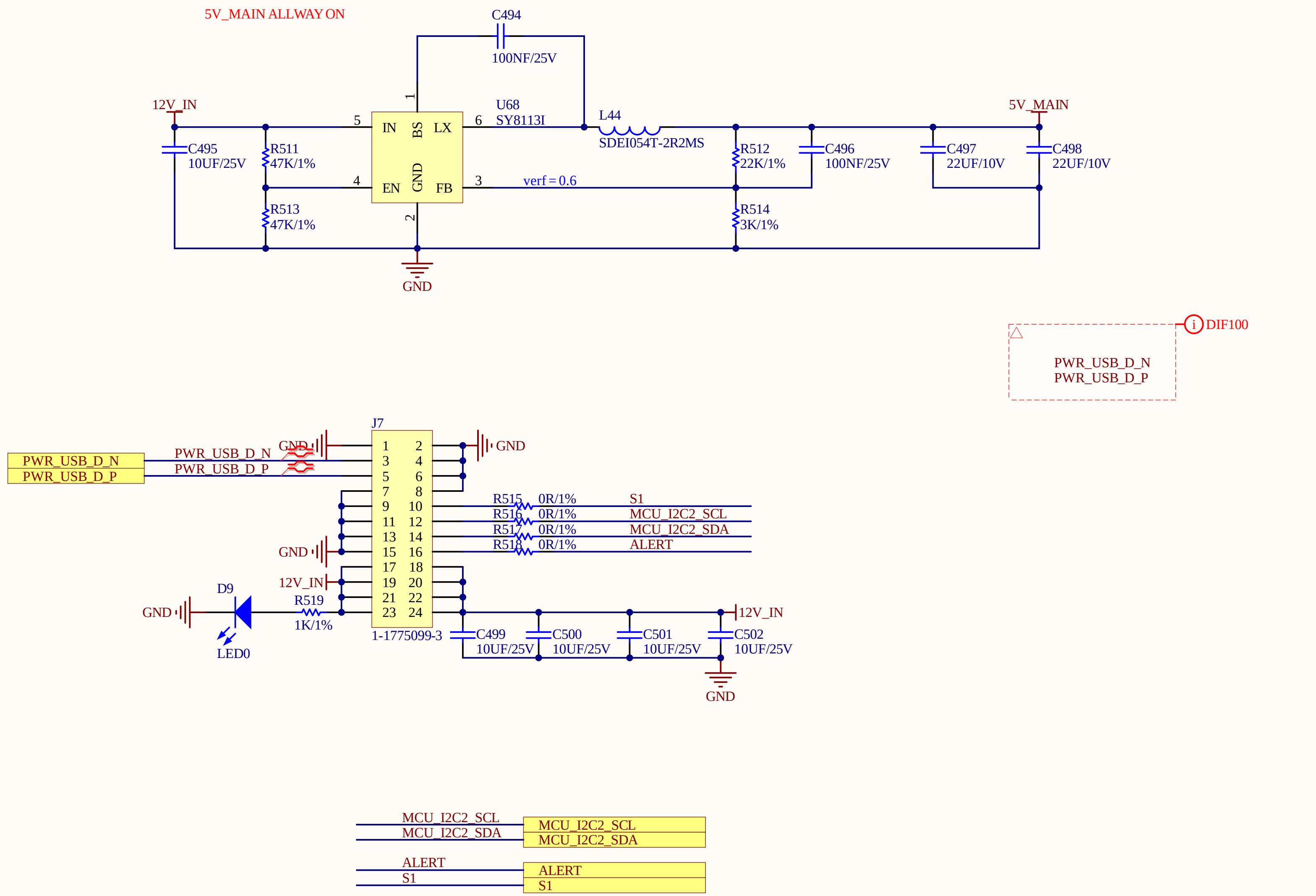
Cannot open file C:\Users\esys\Dropbox\kinte	project: ***HTPSR5G***		**HTP Hitech Co.,Ltd** *OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi*
	Size: A4	Number: 39	Revision: 1.0
	Date: 9/30/2024	Time: 5:16:08 PM	Sheet 39 of 41
	File: CPU CONSOLE.SchDoc		



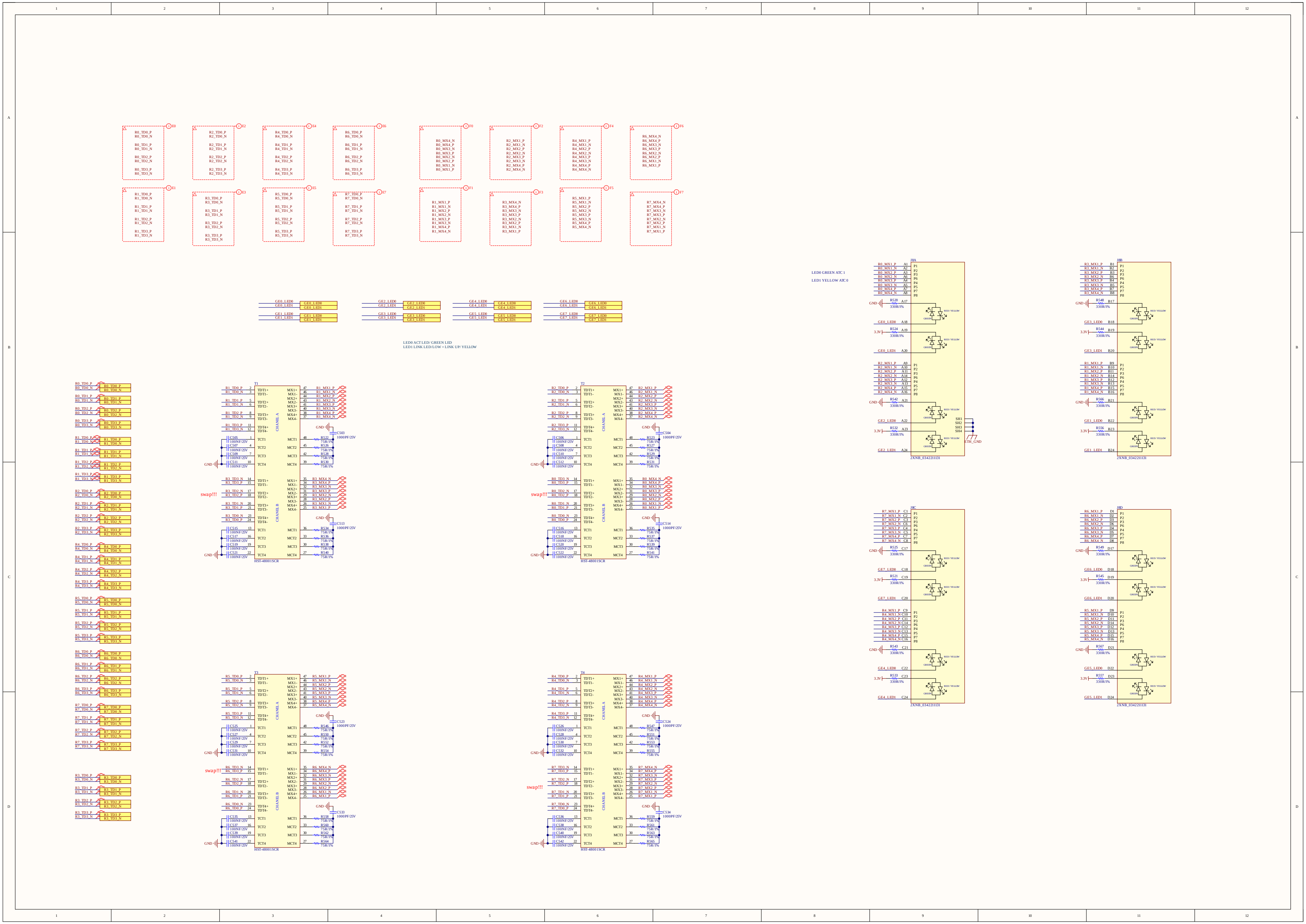
- MCU input include:
+ Done Signal FPGA, need for sense status
+ init signal from FPGA
+ Requet PWR on/off button from external user.
+ PWR good of FPGA pwr
+ PWR good of CPU ATX
+ input data interface from CPU (GPIO OVER USB)
+ input data interface from FPGA (uart is ok)

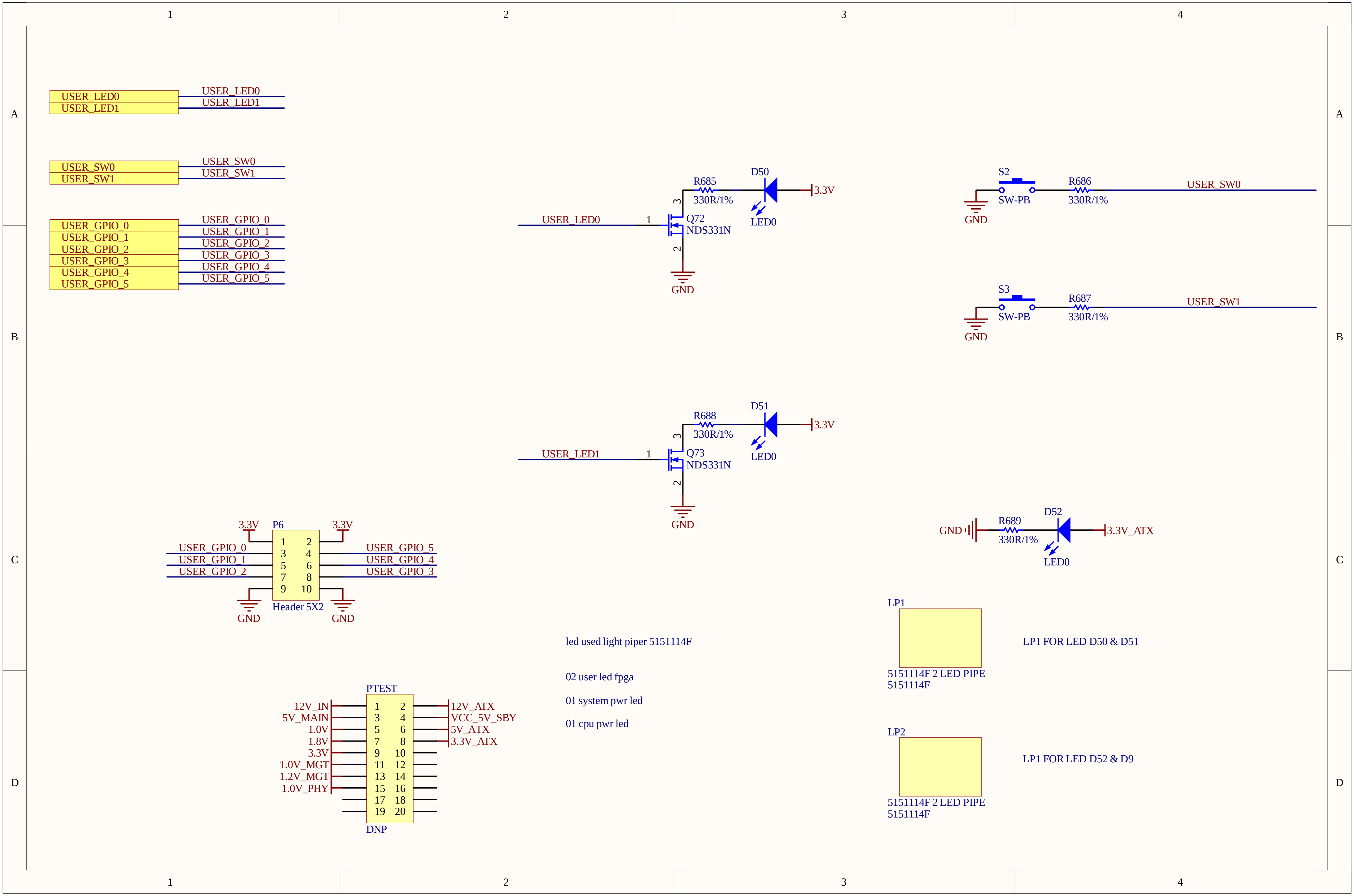
- MCU output include:
+ Enable pwr of FPGA
+ PS_ON of ATX PWR module
+ PWR_button of CPU
+ Reset CPU signal
+ Reset FPGA signal
+ Switch QSPI Flash
+ Control prog signal FPGA
+ FANS control

Cannot open file C:\Users\esys\Dropbox\kinte	project: ***HTPSR5G***		HTP Hitech Co.,Ltd
	Size: A4	Number: 40	Revision: 1.0
	Date: 9/30/2024	Time: 5:16:08 PM	Sheet 40 of 41
	File: SYSTEM CONTROL BLOCK.SchDoc		OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi

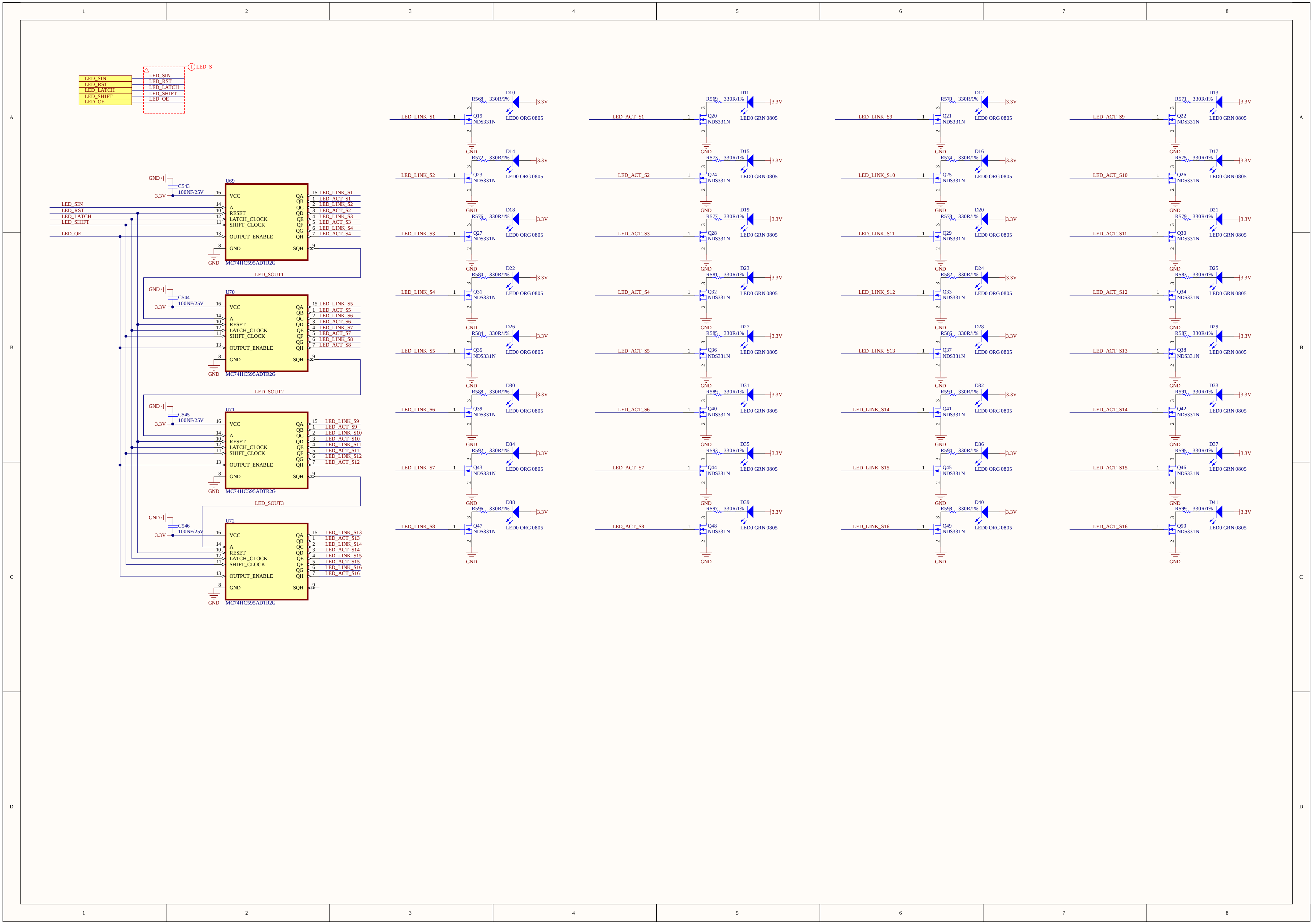


Cannot open file C:\Users\esys\Drop box\kinte	project: ***HTPSR5G***		***HTP Hitech Co.,Ltd***
	Size: A4	Number:41	Revision:1.0
	Date: 9/30/2024	Time: 5:16:08 PM	Sheet41 of 41
	File: SYSTEM PWR.SchDoc		*OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi*

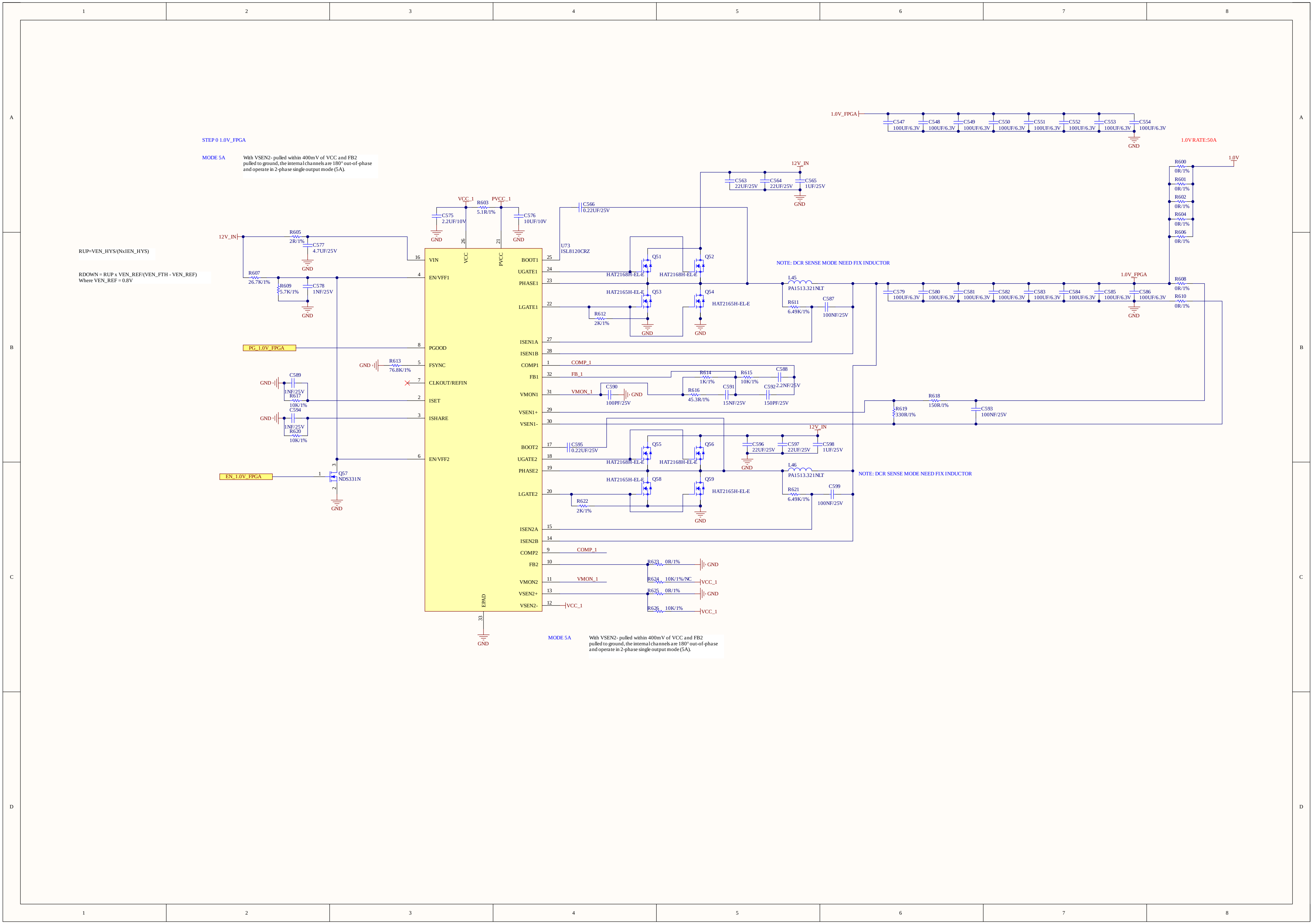

LED0 GREEN ATC 1
LED1 YELLOW ATC 0
LED0 ACT LED/ GREEN LED
LED1 LINK LED/LOW = LINK UP/ YELLOW
J8A
R0_MX1_P A1 P1
R0_MX1_N A2 P2
R0_MX2_P A3 P3
R0_MX2_N A6 P6
R0_MX3_P A4 P4
R0_MX3_N A5 P5
R0_MX4_P A7 P7
R0_MX4_N A8 P8
GND R520 330R/1% A17
RED/ YELLOW
GREEN
GE0_LED0 A18
3.3V R524 330R/1% A19
GE0_LED1 A20
R2_MX1_P A9 P1
R2_MX1_N A10 P2
R2_MX2_P A11 P3
R2_MX2_N A14 P6
R2_MX3_P A12 P4
R2_MX3_N A13 P5
R2_MX4_P A15 P7
R2_MX4_N A16 P8
GND R542 330R/1% A21
GE2_LED0 A22
3.3V R532 330R/1% A23
GE2_LED1 A24
SH1
SH2
SH3
SH4
ETH_GND
2XNB_0342211131
J8B
J8C
J8D
T1
T2
T3
T4
HST-48001SCR
CHANEL A
CHANEL B
swap!!!
1000PF/25V
100NF/25V
75R/1%



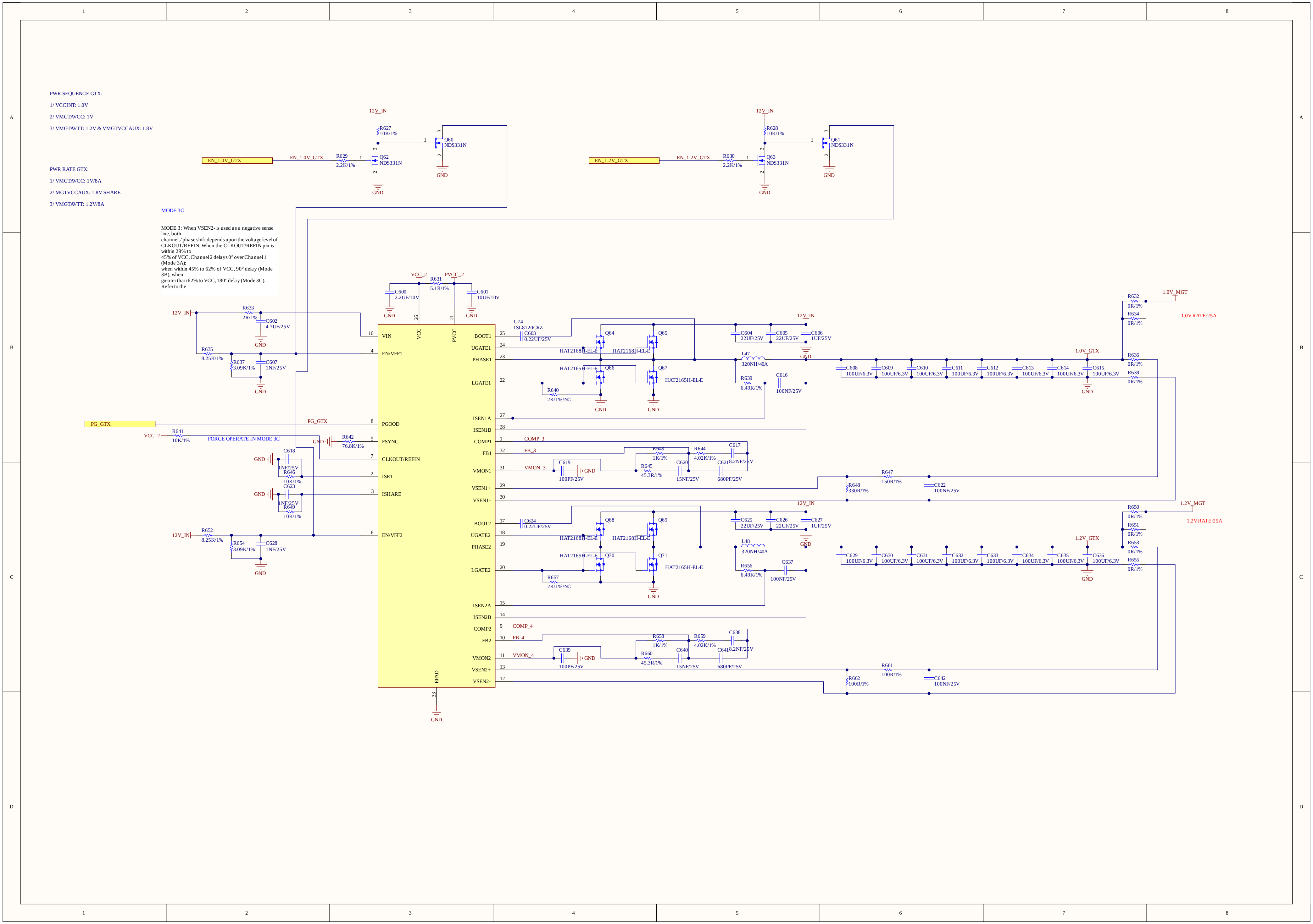
USER_LED0
USER_LED1
USER_SW0
USER_SW1
USER_GPIO_0
USER_GPIO_1
USER_GPIO_2
USER_GPIO_3
USER_GPIO_4
USER_GPIO_5
R685
330R/1%
D50
3.3V
LED0
USER_LED0
Q72
NDS331N
GND
S2
SW-PB
R686
330R/1%
USER_SW0
GND
S3
SW-PB
R687
330R/1%
USER_SW1
GND
R688
330R/1%
D51
3.3V
LED0
USER_LED1
Q73
NDS331N
GND
GND
R689
330R/1%
D52
3.3V_ATX
LED0
3.3V
P6
3.3V
USER_GPIO_0
USER_GPIO_1
USER_GPIO_2
USER_GPIO_5
USER_GPIO_4
USER_GPIO_3
Header 5X2
GND
GND
led used light piper 5151114F
02 user led fpga
01 system pwr led
01 cpu pwr led
LP1
5151114F 2 LED PIPE
5151114F
LP1 FOR LED D50 & D51
LP2
5151114F 2 LED PIPE
5151114F
LP1 FOR LED D52 & D9
PTEST
12V_IN
5V_MAIN
1.0V
1.8V
3.3V
1.0V_MGT
1.2V_MGT
1.0V_PHY
12V_ATX
VCC_5V_SBY
5V_ATX
3.3V_ATX
DNP



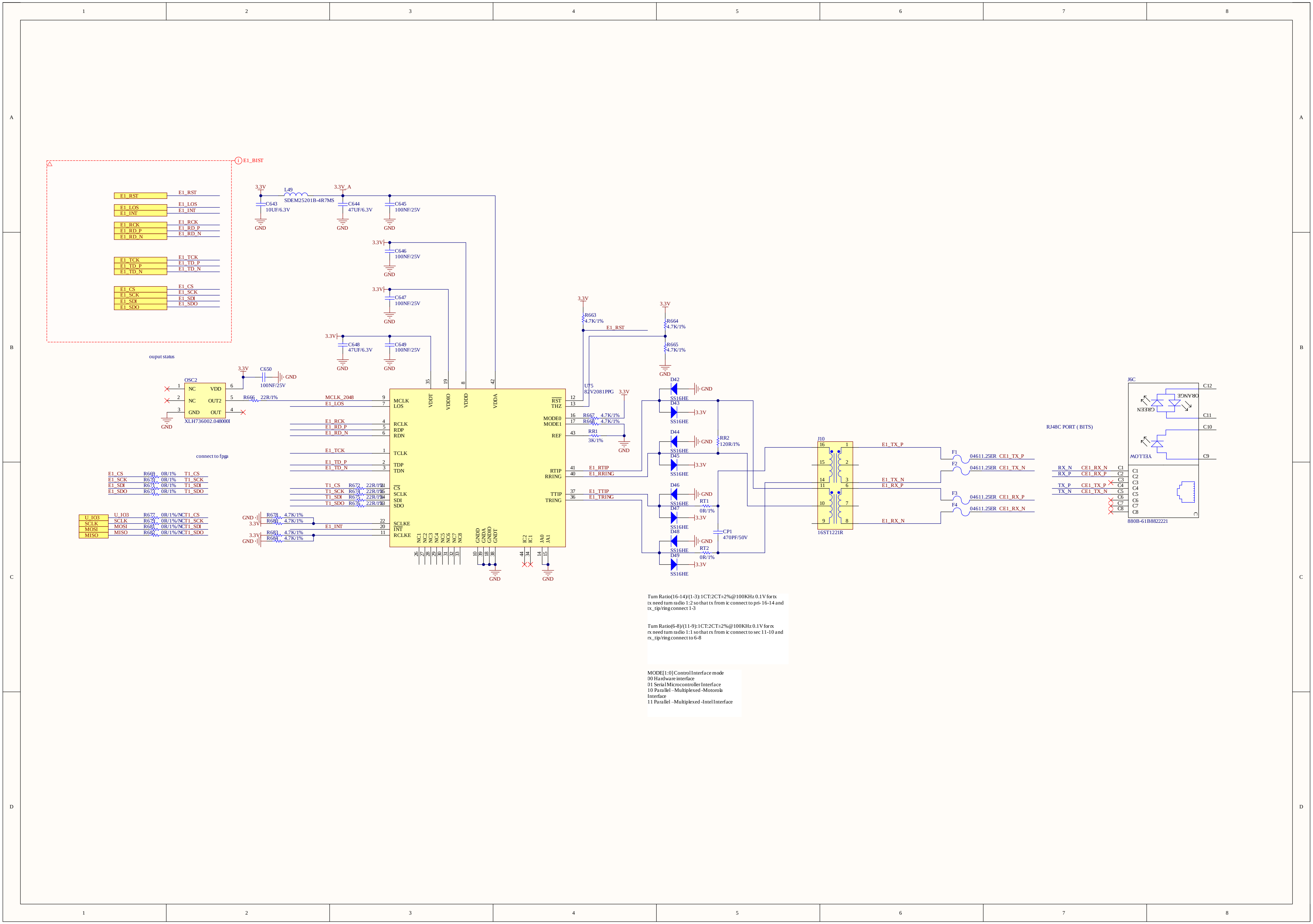
LED_SIN
LED_RST
LED_LATCH
LED_SHIFT
LED_OE
LED_S
GND
3.3V
C543
100NF/25V
U69
VCC
A
RESET
LATCH_CLOCK
SHIFT_CLOCK
OUTPUT_ENABLE
GND
QA
QB
QC
QD
QE
QF
QG
QH
SQH
MC74HC595ADTR2G
LED_LINK_S1
LED_ACT_S1
LED_LINK_S2
LED_ACT_S2
LED_LINK_S3
LED_ACT_S3
LED_LINK_S4
LED_ACT_S4
LED_SOUT1
C544
100NF/25V
U70
LED_LINK_S5
LED_ACT_S5
LED_LINK_S6
LED_ACT_S6
LED_LINK_S7
LED_ACT_S7
LED_LINK_S8
LED_ACT_S8
LED_SOUT2
C545
100NF/25V
U71
LED_LINK_S9
LED_ACT_S9
LED_LINK_S10
LED_ACT_S10
LED_LINK_S11
LED_ACT_S11
LED_LINK_S12
LED_ACT_S12
LED_SOUT3
C546
100NF/25V
U72
LED_LINK_S13
LED_ACT_S13
LED_LINK_S14
LED_ACT_S14
LED_LINK_S15
LED_ACT_S15
LED_LINK_S16
LED_ACT_S16
R568 330R/1% D10 Q19 NDS331N LED0 ORG 0805
R572 330R/1% D14 Q23 NDS331N LED0 ORG 0805
R576 330R/1% D18 Q27 NDS331N LED0 ORG 0805
R580 330R/1% D22 Q31 NDS331N LED0 ORG 0805
R584 330R/1% D26 Q35 NDS331N LED0 ORG 0805
R588 330R/1% D30 Q39 NDS331N LED0 ORG 0805
R592 330R/1% D34 Q43 NDS331N LED0 ORG 0805
R596 330R/1% D38 Q47 NDS331N LED0 ORG 0805
R569 330R/1% D11 Q20 NDS331N LED0 GRN 0805
R573 330R/1% D15 Q24 NDS331N LED0 GRN 0805
R577 330R/1% D19 Q28 NDS331N LED0 GRN 0805
R581 330R/1% D23 Q32 NDS331N LED0 GRN 0805
R585 330R/1% D27 Q36 NDS331N LED0 GRN 0805
R589 330R/1% D31 Q40 NDS331N LED0 GRN 0805
R593 330R/1% D35 Q44 NDS331N LED0 GRN 0805
R597 330R/1% D39 Q48 NDS331N LED0 GRN 0805
R570 330R/1% D12 Q21 NDS331N LED0 ORG 0805
R574 330R/1% D16 Q25 NDS331N LED0 ORG 0805
R578 330R/1% D20 Q29 NDS331N LED0 ORG 0805
R582 330R/1% D24 Q33 NDS331N LED0 ORG 0805
R586 330R/1% D28 Q37 NDS331N LED0 ORG 0805
R590 330R/1% D32 Q41 NDS331N LED0 ORG 0805
R594 330R/1% D36 Q45 NDS331N LED0 ORG 0805
R598 330R/1% D40 Q49 NDS331N LED0 ORG 0805
R571 330R/1% D13 Q22 NDS331N LED0 GRN 0805
R575 330R/1% D17 Q26 NDS331N LED0 GRN 0805
R579 330R/1% D21 Q30 NDS331N LED0 GRN 0805
R583 330R/1% D25 Q34 NDS331N LED0 GRN 0805
R587 330R/1% D29 Q38 NDS331N LED0 GRN 0805
R591 330R/1% D33 Q42 NDS331N LED0 GRN 0805
R595 330R/1% D37 Q46 NDS331N LED0 GRN 0805
R599 330R/1% D41 Q50 NDS331N LED0 GRN 0805



STEP 0 1.0V_FPGA
MODE 5A
With VSEN2- pulled within 400mV of VCC and FB2 pulled to ground, the internal channels are 180° out-of-phase and operate in 2-phase single output mode (5A).
RUP=VEN_HYS/(NxIEN_HYS)
RDOWN = RUP x VEN_REF/(VEN_FTH - VEN_REF)
Where VEN_REF = 0.8V
1.0V_FPGA
1.0V RATE:50A
12V_IN
VCC_1
PVCC_1
U73
ISL8120CRZ
NOTE: DCR SENSE MODE NEED FIX INDUCTOR
L45
PA1513.321NLT
L46
PA1513.321NLT
HAT2168H-EL-E
HAT2165H-EL-E
Q57
NDS331N
PG_1.0V_FPGA
EN_1.0V_FPGA
VIN
EN/VFF1
PGOOD
FSYNC
CLKOUT/REFIN
ISET
ISHARE
EN/VFF2
BOOT1
UGATE1
PHASE1
LGATE1
ISEN1A
ISEN1B
COMP1
FB1
VMON1
VSEN1+
VSEN1-
BOOT2
UGATE2
PHASE2
LGATE2
ISEN2A
ISEN2B
COMP2
FB2
VMON2
VSEN2+
VSEN2-
EPAD
COMP_1
FB_1
VMON_1
GND
1.0V
MODE 5A
With VSEN2- pulled within 400mV of VCC and FB2 pulled to ground, the internal channels are 180° out-of-phase and operate in 2-phase single output mode (5A).



PWR SEQUENCE GTX:
1/ VCCINT: 1.0V
2/ VMGTAVCC: 1V
3/ VMGTAVTT: 1.2V & VMGTVCCAUX: 1.8V
PWR RATE GTX:
1/ VMGTAVCC: 1V/8A
2/ MGTVCCAUX: 1.8V SHARE
3/ VMGTAVTT: 1.2V/8A
MODE 3C
MODE 3: When VSEN2- is used as a negative sense line, both
channels' phase shift depends upon the voltage level of CLKOUT/REFIN. When the CLKOUT/REFIN pin is within 29% to
45% of VCC, Channel 2 delays 0° over Channel 1 (Mode 3A);
when within 45% to 62% of VCC, 90° delay (Mode 3B); when
greater than 62% to VCC, 180° delay (Mode 3C).
Refer to the
EN_1.0V_GTX
EN_1.2V_GTX
PG_GTX
12V_IN
VCC_2
PVCC_2
GND
R627 10K/1%
R628 10K/1%
R629 2.2K/1%
R630 2.2K/1%
Q60 NDS331N
Q61 NDS331N
Q62 NDS331N
Q63 NDS331N
U74 ISL8120CRZ
VIN
EN/VFF1
PGOOD
FSYNC
CLKOUT/REFIN
ISET
ISHARE
EN/VFF2
VCC
PVCC
BOOT1
UGATE1
PHASE1
LGATE1
ISEN1A
ISEN1B
COMP1
FB1
VMON1
VSEN1+
VSEN1-
BOOT2
UGATE2
PHASE2
LGATE2
ISEN2A
ISEN2B
COMP2
FB2
VMON2
VSEN2+
VSEN2-
EPAD
C600 2.2UF/10V
R631 5.1R/1%
C601 10UF/10V
R633 2R/1%
C602 4.7UF/25V
R635 8.25K/1%
R637 3.09K/1%
C607 1NF/25V
R641 10K/1%
FORCE OPERATE IN MODE 3C
R642 76.8K/1%
C618 1NF/25V
R646 10K/1%
C623 1NF/25V
R649 10K/1%
R652 8.25K/1%
R654 3.09K/1%
C628 1NF/25V
C603 0.22UF/25V
C624 0.22UF/25V
Q64 Q65 Q66 Q67 Q68 Q69 Q70 Q71
HAT2168H-EL-E
HAT2165H-EL-E
R640 2K/1%/NC
R657 2K/1%/NC
C604 22UF/25V
C605 22UF/25V
C606 1UF/25V
C625 22UF/25V
C626 22UF/25V
C627 1UF/25V
L47 320NH/40A
L48 320NH/40A
R639 6.49K/1%
R656 6.49K/1%
C616 100NF/25V
C637 100NF/25V
C608 100UF/6.3V
C609 100UF/6.3V
C610 100UF/6.3V
C611 100UF/6.3V
C612 100UF/6.3V
C613 100UF/6.3V
C614 100UF/6.3V
C615 100UF/6.3V
C629 100UF/6.3V
C630 100UF/6.3V
C631 100UF/6.3V
C632 100UF/6.3V
C633 100UF/6.3V
C634 100UF/6.3V
C635 100UF/6.3V
C636 100UF/6.3V
1.0V_GTX
1.2V_GTX
1.0V_MGT
1.2V_MGT
1.0V RATE:25A
1.2V RATE:25A
R632 0R/1%
R634 0R/1%
R636 0R/1%
R638 0R/1%
R650 0R/1%
R651 0R/1%
R653 0R/1%
R655 0R/1%
COMP_3
FB_3
VMON_3
COMP_4
FB_4
VMON_4
R643 1K/1%
R644 4.02K/1%
C617 8.2NF/25V
R645 45.3R/1%
C619 100PF/25V
C620 15NF/25V
C621 680PF/25V
R647 150R/1%
R648 330R/1%
C622 100NF/25V
R658 1K/1%
R659 4.02K/1%
C638 8.2NF/25V
R660 45.3R/1%
C639 100PF/25V
C640 15NF/25V
C641 680PF/25V
R661 100R/1%
R662 100R/1%
C642 100NF/25V



Turn Ratio(16-14)/(1-3):1CT:2CT±2%@100KHz 0.1V for tx
tx need turn radio 1:2 so that tx from ic connect to pri-16-14 and tx_tip/ring connect 1-3

Turn Ratio(6-8)/(11-9):1CT:2CT±2%@100KHz 0.1V for tx
rx need turn radio 1:1 so that rx from ic connect to sec 11-10 and rx_tip/ring connect to 6-8

MODE[1:0] Control Interface mode
00 Hardware interface
01 Serial Microcontroller Interface
10 Parallel –Multiplexed -Motorola Interface
11 Parallel –Multiplexed -Intel Interface